



MANufacturing as a Service And supply chain predictIVE TWIN for critical raw materials

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D6.1: Communication, Dissemination, Engagement strategic plan

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Contents

Version History.....	2
Contents.....	3
List of Figures.....	4
List of Tables.....	5
Abstract.....	6
Glossary.....	7
1. Introduction	8
1.1. Deliverable structure.....	8
2. Dissemination and communication strategy.....	8
2.1. Objectives of the dissemination and communication plan	10
2.2. Identified target audiences.....	11
2.3. Timeline for dissemination and communication activities	13
3. Dissemination and communication tools, channels and materials	13
3.1. Visual identity of the project.....	14
3.1.1. Logo and colours.....	14
3.1.2. PowerPoint presentation template.....	15
3.2. Project website	15
3.3. Press Release	17
3.4. Newsletters.....	17
3.5. Printed Dissemination and Communication material	18
3.5.1. Brochure and Leaflet.....	18
3.5.2. Poster	18
3.6. Introductory presentation of the project.....	18
3.7. Researchers' introduction	19
3.8. Project videos	19
3.9. Social media	19
3.9.1. LinkedIn	20
3.9.2. X (formerly Twitter).....	20
3.10. Industrial fairs/exhibitions, scientific conferences and publications	21
3.11. Thematic workshops	23
4. Dissemination reporting, monitoring and evaluation	25
4.1. Monitoring and reporting of performed activities.....	25

4.2. Evaluation of performed activities	26
4.3. Key performance indicators	27
5. Networking	27
6. Appendices	31
Annexe 1 – PowerPoint Presentation template.....	31
Annexe 2 – Project website images.....	33
Annexe 3 – 1st Press Release	38
Annexe 4 – Brochure.....	39
Annexe 5 – Leaflet	40
Annexe 6 – Poster.....	41
Annexe 7 – General Introductory Presentation	41
Annexe 8 – Events, Journals, Magazines lists.....	43
Annexe 9 – Dissemination, Communication and Publications reporting.....	45

List of Figures

Figure 1: Communication, Dissemination, Exploitation (source: https://data.europa.eu/doi/10.2848/289075)	9
Figure 2: MaaSiveTwin logo	15
Figure 3: MaaSiveTwin colour palette (hex codes)	15
Figure 4: MaaSiveTwin LinkedIn profile	20
Figure 5: MaaSiveTwin X profile	21
Figure 6: MaaSiveTwin presentation template - Title slide.....	31
Figure 7: MaaSiveTwin presentation template - Main slide.....	31
Figure 8: MaaSiveTwin presentation template - Closing slide.....	32
Figure 9: MaaSiveTwin website – Homepage.....	33
Figure 10: MaaSiveTwin website – Project tab; Key Objectives & Specific Objectives	34
Figure 11: MaaSiveTwin website – Project tab; Key elements of Methodology.....	34
Figure 12: MaaSiveTwin website – Project tab; Methodology concept (interactive element).35	
Figure 13: MaaSiveTwin website – Consortium tab	35
Figure 14: MaaSiveTwin website – Public Documents tab.....	36
Figure 15: MaaSiveTwin website – News tab	36
Figure 16: MaaSiveTwin website – Contact Us tab.....	37
Figure 17: MaaSiveTwin website – Main Menu	37
Figure 18: Press Release in ENG.....	38
Figure 19: Brochure.....	39
Figure 20: Leaflet	40
Figure 21: Poster A0	41

Figure 22: Comprehensive list of scientific or industrial events (conferences, exhibitions, fares, etc.) of interest to MaaSiveTwin.....	43
Figure 23: Comprehensive list of Scientific Journals and Industrial Magazines of interest to MaaSiveTwin.....	44
Figure 24: Communication reporting via SharePoint - new activity form	45
Figure 25: Dissemination reporting via SharePoint - new activity form.....	46
Figure 26: Publications reporting via SharePoint - new publication form.....	47

List of Tables

Table 1: Objectives of the dissemination activities.....	11
Table 2: MaaSiveTwin Stakeholders	12
Table 3: Events-Journals-Magazines of interest to MaaSiveTwin	23
Table 4: Effectiveness ranking for dissemination and communication activities depending on the tool used.....	26
Table 5: Key Performance Indicators for dissemination and communication activities.....	27
Table 6: List of relevant projects	30

Abstract

Dissemination and communication activities are of vital importance for the project in order to maximise the impact and trigger engagement across the targeted groups and stakeholders. Communication activities serve as the basis of dissemination activities for MaaSiveTwin. It is thus crucial to work on them continuously throughout the project.

The purpose of this document is to present a Communication, Dissemination and Engagement strategic plan for the MaaSiveTwin project and to ensure the maximum impact of the project during its lifespan.

This Plan (D6.1) was prepared at the beginning of the project and describes the overall dissemination and communication goals, target groups, dissemination channels, reporting, monitoring and evaluation procedures, etc. The dissemination and communication plan will provide a regular flow of information, thus contributing to the project's visibility, and will increase opportunities to reach out to the public.

The Plan also describes plans for networking activities with other relevant projects, initiatives and platforms.

This Plan has been drafted considering all the dissemination and communication aspects mentioned in the Grant Agreement and Consortium Agreement. Dissemination achievements will be reported to the European Commission based on the provisions of the Grant Agreement.

Glossary

Term	Explanation
CA	Consortium Agreement
CRM	Critical Raw Materials
DF	Data Fusion
GA	Grant Agreement
IP	Intellectual Property
IPR	Intellectual Property Rights
KER	Key Exploitable Result
KPI	Key Performance Indicator
ML	Machine Learning
R&D	Research & Development
R&I	Research & Innovation

1. Introduction

Deliverable “D6.1 - Communication, Dissemination, Engagement strategic plan” was developed in the context of “Task 6.1 - Communication, dissemination and engagement plan” under “WP6 - Communication, Dissemination, and Exploitation”. The purpose of this deliverable is to consolidate, from day one, the overall engagement strategy of the MaaSiveTwin project, to establish the objectives of the distribution and communication activities, to determine the most successful ways of achieving them, and to break them down into a comprehensive implementation action plan. To this end, the deliverable sets out the aims, instruments, resources, and platforms to be used to efficiently distribute the events, milestones and tangible outcomes of the project according to the targeted audiences. This deliverable also aims to develop clear KPIs of the planned activities based on which monitoring and assessment will be carried out.

1.1. Deliverable structure

The current deliverable contains an overview of the dissemination and communication objectives, the identified target audiences as well as the timeline of the planned activities (Section 2). Section 3 describes the dissemination and communication tools, channels and materials (including the project logo, project presentation template, project website, press releases, newsletters, printed dissemination and communication materials and the social media presence of the MaaSiveTwin project). Dissemination reporting, monitoring and evaluation, including Key Performance Indicators (KPIs) set by the project partners, are discussed in Section 4. Section 5 provides an overview of the planned networking activities with other EU-funded projects and relevant initiatives.

The deliverable also contains 9 Annexes providing images about the already available dissemination materials and tools created for effectively implementing this plan.

2. Dissemination and communication strategy

As shown in Figure 1 below, communication is promoting the project itself and its results with the aim of reaching multiple audiences, while dissemination means making project results public via Open Science. While communication activities aim to engage with stakeholders and show the success of European collaboration, dissemination actions target to maximise results' impact and make scientific results a common good.

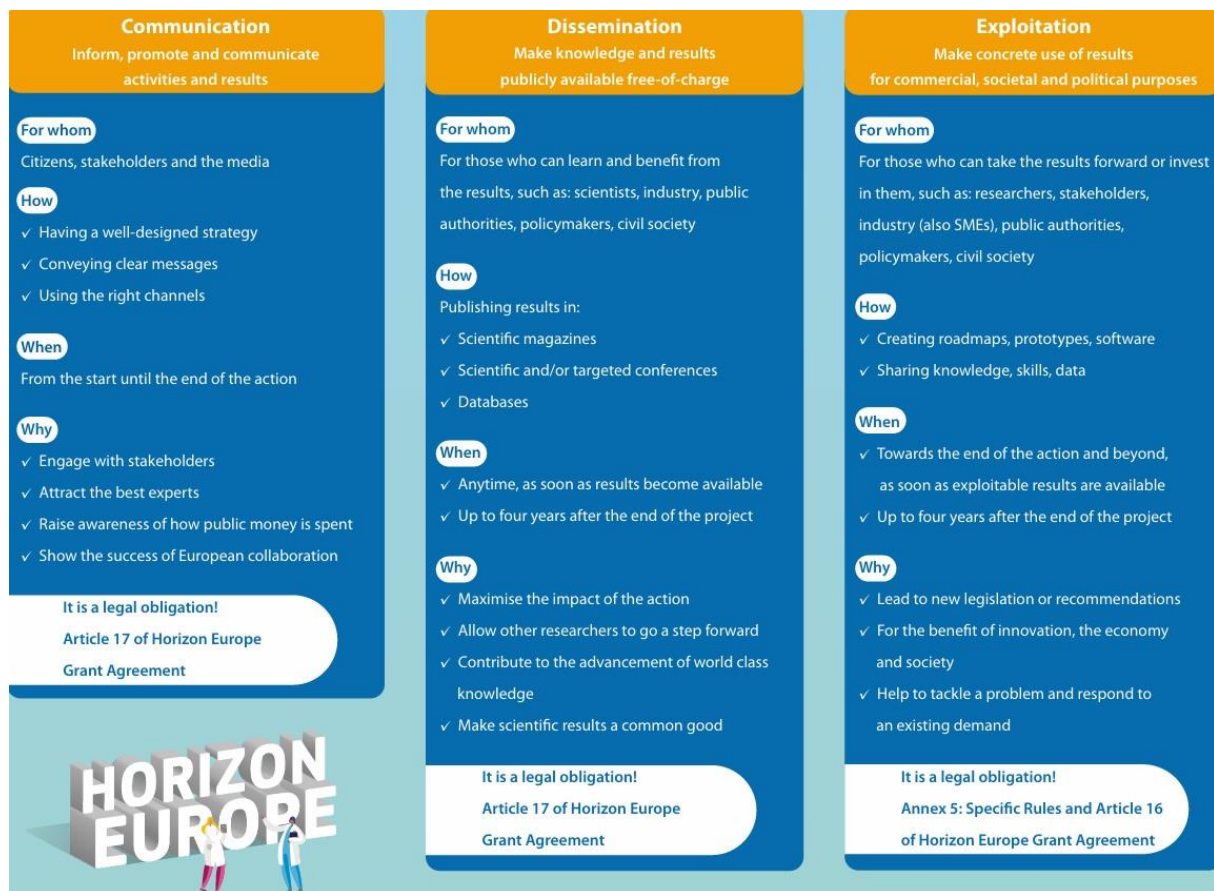


Figure 1: Communication, Dissemination, Exploitation (source: <https://data.europa.eu/doi/10.2848/289075>)

In its dissemination strategy, the MaaSiveTwin project focuses on raising awareness of actual and potential stakeholders. Any organisation, individual or business with a vested interest in the project or will be influenced by its results is considered a stakeholder. In terms of interest in the project, MaaSiveTwin focuses on recognising and evaluating these stakeholders before engaging them. This information is considered essential for the progress of the project and the direction of further dissemination activities.

Dissemination is defined as the means through which research results are presented to the public. The process must take place through close cooperation between project partners. To produce and transmit knowledge on innovation, reaching a common understanding of the means and the key messages to communicate is crucial. Dissemination is usually not a one-way operation, and by including the overall awareness and visibility of the project, this term suggests a broader definition. To be successful, dissemination should start raising awareness before any results are published.

The following principles govern the current Dissemination and Communication Plan:

- The communication processes must be transparent and known to all consortium partners.
- Dissemination and communication must be purposeful and timely.
- Dissemination and communication must be open and honest.

Communication is a two-way process: it is not just a matter of conveying messages but also receiving feedback and starting a meaningful dialogue with relevant stakeholders. Additionally, horizontal and vertical communication ways are equally important. Effective communication contains different elements, such as:

- Deliver information.
- Information received by the audience.
- Information understood by the audience.
- The audience takes the information and shares it with someone else.
- The audience is convinced of the information received.

MaaSiveTwin is adopting a funnelled approach, similar to a marketing funnel, to ensure broad but targeted communication within its ecosystem, enabling active engagement and achieving efficient dissemination of project results. Such an approach will primarily focus on raising awareness of the targeted stakeholders about key aspects and benefits of the MaaSiveTwin project results. Easy-to-understand visual materials will be used to make concepts and benefits instantly recognisable to a broad audience. This aims to cultivate further interest in potential end-users who will be directed to more detailed information and material about the developed technology. Customised content will be communicated to specific target audiences, aiming to create and maintain an active ecosystem for stakeholders. Similarly, relevant information will be extracted from project deliverables, interviews with partners, industry reports, etc., and relayed through the MaaSiveTwin communication channels to further support active engagement.

2.1. Objectives of the dissemination and communication plan

The aim of the dissemination strategy is to distribute knowledge about the project outputs, innovation potential, and business opportunities to all relevant stakeholders. The dissemination strategy must be adapted to present relevant information at the right time, place and context to allow for the largest diffusion.

The main objectives of the dissemination and communication action plan are:

- Guarantee an effective communication of the project messages and activities at the local, national, and EU levels.
- Identify appropriate target audiences to convey the dissemination and communication messages.
- Implement a wide and differentiated set of dissemination tools and events.
- Identify the dissemination and communication KPIs, necessary to measure the effectiveness and efficiency of the performed activities.
- Define how the project will cooperate with other EU-funded projects or related initiatives.
- Define how the dissemination activities will be reported on.

The objectives of the dissemination activities include setting up a strategy for the main tasks that need to be addressed and decisions that belong to each task:

Plan	The first stage begins by identifying target audiences, messages, tools, and channels. Based on these findings, an adequate and effective communication and dissemination plan has to be built to maximise the impact of project results.
Design	When producing dissemination tools, it will be necessary to design a comprehensive set of communication materials (both printed and online) to ensure an easy identification and a significant exposure of the project.
Distribute	Dissemination channels need to be identified, and dissemination materials must be adapted to get the full effect of using the channels.
Participate	Project events must be organised, and participation in workshops, conferences, and meetings will be encouraged.
Report	Collect dissemination and communication reports from project partners via a dedicated form available on SharePoint. These reports will serve as the basis for the monitoring and evaluating performed activities.
Monitor and evaluate	Performed activities and connected KPIs will be monitored and benchmarked towards the KPIs established in this deliverable. The current Dissemination and Communication Plan will be continuously updated based on the evaluation outcome of performed activities.
Review and improve	Ensure persistent and long-lasting visibility of the project activities and outcomes. Review the achieved visibility and improve the effectiveness of dissemination and communication activities.

Table 1: Objectives of the dissemination activities

The project's dissemination and communication activities will be directly linked to the exploitation of the project results. Efficient publicity and wide visibility of the project and its achievements will increase stakeholders' engagement, hence using project results beyond the project's lifetime. Ultimately, the communication and dissemination activities will maximise the project's impact.

2.2. Identified target audiences

MaaSiveTwin project partners have already identified the major stakeholders and target audiences. These include on one hand scientific, academic, research organisations in order to facilitate discussion and exchange of information on the scientific and technological advancement in the field of supply chain management of Critical Raw Materials (CRMs) and on the other hand industrial organisations with the aim to facilitate and enhance future exploitation and commercialisation activities of the MaaSiveTwin solution. Additionally, relevant other EU initiatives and EU funded projects are also among the targeted groups (for more information on networking activities with these initiatives and projects are given in Section 5). Table 2 below provides a list of identified stakeholders for the MaaSiveTwin project.

Stakeholder	Strategic objective/expectations
Manufacturers along the CRM and rechargeable battery value chain, magnets, VRFB and fuel cell manufacturers that include: - mining companies - refinery companies	• Data platform for making supply chain tracking for CRMs. • Early adopters for MaaSiveTwin

- intermediate (precursor) product producers - rechargeable battery producers - green energy parts suppliers - electric vehicle producers - energy storage system manufacturers - renewable energy producers - recycling companies	
Procurement and analysts	Research and develop models to predict CRM supply/demand/price.
EU institutions	Enable a resilient production of CRM value chains.
Research institutes	Provide data and innovative methods for EU R&D activities.
NGOs associations	Showcase MaaSiveTwin solutions as best practices.
Policymakers/standardisation bodies (EU and national level)	• Recommendations for the design of the MaaS platform and good practices to fully exploit CRM value chain as well as the project's contribution to the Critical Raw Materials Act's goals. • Early adopters for MaaSiveTwin
General public/consumers	Create awareness on the clean sourced CRMs and the project's impact on creating a competitive and sustainable digital Europe and green value chain.
Other EU-funded projects working on the same topic	Create synergies, share R&I results and confront them with the scientific community

Table 2: MaaSiveTwin Stakeholders

Besides the target groups mentioned in Table 2, MaaSiveTwin will seek collaboration with initiatives, associations and clusters, with the aim of facilitating the creation of a new value chain for the future exploitation of project results. They will also act as multipliers for the dissemination of project results. The following initiatives, associations and clusters will be approached:

- **EU Green Deal:** is a growth strategy that protects the climate, based on a clean and circular economy, which has a vision for Europe to remain a prime destination for investments that bring stable, future-proof quality jobs with a strong industrial base.
- **EU Battery Passport:** is a core tool that enables globally sustainable scaling of battery value chains. Battery Passport is a lighthouse example of using innovation to achieve the overarching goals of the EU Green Deal, decarbonisation and dematerialisation while supporting the legal implementation and empowering companies to make informed decisions on their supply chains and products.
- **European Battery Alliance – EU EBA250:** a unique platform for key stakeholders throughout the entire battery value chain. The network includes the European

Commission, EU member states, the European Investment Bank and more than 800 industrial, innovation and academia stakeholders. The objective is to build a strong pan-European battery industry that is able to help Europe capture a growing market worth 250B€/year from 2025.

- **Important Project of Common European Interest – IPCEIs on battery value chain:** a two-part IPCEI has been implemented to promote battery production. The IPCEI on Batteries and the IPCEI European Battery Innovation (EuBatIn). Both IPCEIs have in common that their participants represent the complete value chain, from material through the cells to the battery system and the final step of recycling. At the same time, there is a high degree of networking between the companies themselves and the two IPCEIs.
- **Batteries European Partnership – BATT4EU:** is a Co-programmed Partnership established under Horizon Europe, the Framework Programme for Research and Innovation of the European Union. BATT4EU is a collaboration of the European Commission and European battery R&I stakeholders, united in BEPA that aims to achieve a competitive and sustainable European industrial value-chain for e-mobility and stationary applications.
- **Rare Earth Industry Association – REIA:** is an international non-profit organisation representing the global REE value chain, including SMEs, global corporates, academia, governments and NGOs.
- **EIT Raw Materials – EIT:** Dedicated on developing raw materials into a major strength for Europe by boosting competitiveness, growth and attractiveness of the European raw materials sector via radical innovation, new education approaches and guided entrepreneurship.

2.3. Timeline for dissemination and communication activities

- **Planning phase:** Months 1-6 of the project; set up dissemination and communication action plan.
- **Implementation phase:** Months 1-48 of the project: produce a comprehensive set of tools and materials, perform dissemination and communication activities, and reach out to stakeholders.
- **Monitoring and evaluation:** Months 2-48: monitor and evaluate the efficiency and impact of the performed dissemination and communication activities based on continuous reporting and by benchmarking the impact achieved to the pre-established KPIs.

3. Dissemination and communication tools, channels and materials

The variety of target groups demands the use of several channels for the MaaSiveTwin project to inform, communicate and engage stakeholders effectively. To reach this wide audience, MaaSiveTwin uses a mix of traditional and online communication channels. Online channels include the project website, social media and bi-annual electronic newsletters, whereas the

traditional channels cover conferences, events, exhibitions, trade fairs, workshops, press appearances, seminars, etc. Some have a one-directional scope (aimed at informing the target audience) and the other a bi-directional scope (aimed at triggering dialogue and ensuring active stakeholder involvement).

3.1. Visual identity of the project

3.1.1. Logo and colours

The design of the MaaSiveTwin project logo is well-justified through its thoughtful visual and conceptual elements. The colour scheme of pink and purple on the logo symbolises, on the one hand, the colour of several rare earth elements and CRMs that can exhibit under certain conditions (i.e., lithium when burned can produce a pink flame, neodymium when oxidized, nickel in hydrated forms, manganese sulphate when complexed with other ligands, etc.). On the other hand, purple symbolises leadership and supports inventiveness, thus aligning perfectly with the project's objectives and reflecting the project's scope of developing an innovative solution that will make a substantial impact by addressing challenges in the CRM value chain, fostering a more efficient, resilient and substantial industrial ecosystem. Additionally, the display of the accumulated small powder on the logo represents the powder pile of raw material powders and reflects the raw data and huge amount of information that will be gathered on the battery-related supply chain as well as events (i.e. flood, blockage, policy) that disrupt the supply chain. The slightly distorted/compacted reflection of the letters symbolises how the challenge will be solved within the project by consolidating different information together and applying a reasonable and logical model, therefore the total output is slightly distorted from the raw data. Finally, the word "Twin" represents the digital twin platform, which is MaaSiveTwin Project's solution that will use machine learning algorithms to process large quantities of sensor data and identify data patterns, subsequently enabling the key players dependent on CRM to make early decisions to address supply chain disruptions promptly.

Developing a visual identity and a logo ensures that project outputs are consistent and easily recognisable. The project logo will be used in the following cases:

- All documents developed within the framework of the project, including those submitted to the EC (e.g., deliverables).
- Presentations that will be used for dissemination and communication activities to be carried out by each participant within the framework of the project.
- MaaSiveTwin website and social media accounts.
- Printed dissemination and communication materials.

The project logo was conceptualised by Battronics and re-designed in high resolution by RTD Talos at the beginning of the project.

The following logo format should always be used in the correct dimensions to avoid distortion and ensure the consistency needed for effective brand recognition.



Figure 2: MaaSiveTwin logo

The MaaSiveTwin colour palette is meticulously crafted for optimal performance in digital environments. The vibrant and contemporary colours provide striking impact and contrast to our digital products.

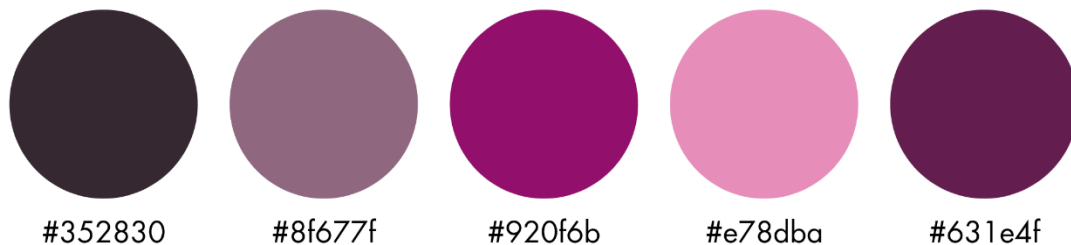


Figure 3: MaaSiveTwin colour palette (hex codes)

3.1.2. PowerPoint presentation template

A PowerPoint presentation template containing the MaaSiveTwin project logo, partner logos, SERI funding acknowledgement and its emblem, and Horizon Europe funding acknowledgement, together with the EU emblem, was also created to be used by all project partners. The presentation template ensures a unified appearance and easy identification of the project by the stakeholders and the general public. The presentation template is given in Annexe 1.

3.2. Project website

The project website (<https://maasivetwinproject.eu>) is one of the main dissemination and communication tools as it has the capability to address a wide range of stakeholders by providing easy and quick access to the information related to the project. It is the central point of information provision and links to other communication channels, such as LinkedIn and X (formerly Twitter). It contains the most crucial information about the project and the consortium, and it is maintained and updated regularly to promote and disseminate project activities, results and promotional outputs.

The website was developed user-friendly, allowing the visitors to get an overview of the project, its objectives, processes, and consortium partners. It was designed to be compatible across devices (PCs, mobile phones, tablets) while also providing an interactive experience, attracting users to engage and learn about all aspects of MaaSiveTwin. The website provides information about the project's challenge, objectives, methodology, modelling techniques, proposed solution as well as impact on the CRM value chain. A "Contact Us" tab can also be

found on the website, where the project email (info@maasivetwinproject.eu) is provided to provide viewers with a means to reach out for additional information as well as for stakeholders and other projects to be able to easily communicate with the project for potential collaborations and joint activities. A newsletter form is also provided, which the audiences interested in the project can fill out and, therefore, subscribe to and receive every electronic newsletter of the project in their email. All content is written and couched in plain English language, ensuring a broad audience can easily understand the key messages. This approach promotes inclusivity and encourages engagement from diverse stakeholders, making the information more approachable for the general public while maintaining clarity for specialists.

The website is composed of the following sections (images are provided in Annexe 2):

1. Homepage:
 - a. Challenge
 - b. Solution
 - c. Impact
 - d. MaaSiveTwin Fast Facts
2. Project
 - a. Key Objectives
 - b. Specific Objectives
 - c. Key Elements of Methodology
 - d. Methodology Concept
3. Consortium
4. Public Documents
 - a. Communication Materials
 - b. Public Deliverables
 - c. Publications
 - d. Videos
5. News
6. Contact Us

The following statements appear in the footer of all pages:

- Newsletter subscription form.
- Contact email of project (info@maasivetwinproject.eu).
- Acknowledgements of EU and SERI, together with their emblems and the Grant Agreement Number and Contract number, respectively.
- Links to social media (LinkedIn, X).

Management of the website:

- Technical management: Talos, as Work Package and task leader, is responsible for the technical management of the website. Technical management is implemented through the WordPress Content Management System, which allows the users to change the content on their website without necessarily having knowledge of web development. Therefore, it ensures that the website content can be easily updated to prevent any unnecessary delay in uploading files, texts, images or other assets.

- Management of the website content: as far as the content is concerned, any document, picture, icon or text inserted into the webpage must first be categorised as “non-confidential” to ensure that any potential future protection of the project results will not be jeopardised and to prevent infringement of partners’ intellectual property rights. Talos is responsible for collecting the necessary materials from the partners, circulating them within the consortium, and obtaining partners’ consent to publish them on the website.

3.3. Press Release

The first press release of the MaaSiveTwin project was issued right after the project started, informing the general audiences, target groups and all stakeholders about the project launch. The press release contains a general description of the project and the consortium. The press release was uploaded to the project website in the Public Documents tab under the Communication Materials section and distributed through social media channels and partners’ networks. Press releases will be prepared and published annually in four different languages. The primary language will be English, and the other three languages will alternate each year, which will be the mother languages of each partner (German, Greek, Polish, Swiss, Serbian). The first issue of the project is presented in Annexe 3 of this deliverable in all four languages (English, German, Greek, and Polish).

3.4. Newsletters

Newsletters will be prepared, published, and distributed to cover the progress of the last six months of the project, providing the latest information about the project developments and other related subjects. A total number of nine newsletters will be prepared until the end of the project.

The first newsletter will be published in early September 2024 (month 6), covering the following topics:

- Cover page with a short introduction to the project
- The Challenge and Solution
- Project objectives
- Expected impact
- Methodology concept
- Short consortium introduction
- Short interview with the project coordinator
- Latest news and project activities
- Contact information (social media links)

The newsletter will be distributed electronically:

- via the project website (in both “Public Documents” and “News” tabs).
- via the project’s social media channels (LinkedIn, X).
- via partners’ communication channels.
- via email to the newsletter subscribed contact list.

3.5. Printed Dissemination and Communication material

3.5.1. Brochure and Leaflet

The project brochure (bi-fold design, presented in Annexe 4) and project leaflet (tri-fold design, presented in Annexe 5) provide information about the MaaSiveTwin project. More specifically:

The brochure (general information about the project) contains:

- Project general information (fast facts).
- Project introduction (containing the challenge, excellence, and solution).
- Innovation Pillars of the project.
- Methodology concept.
- Consortium.
- Social media accounts (QR codes format with their link embedded).
- Acknowledgments of funding agencies (EU & SERI).

The leaflet (more specific information about the project) contains:

- Challenge.
- Solution.
- Methodology concept.
- Project's impact.
- Specific objectives.
- Key elements of methodology.
- Consortium.
- Social media accounts (QR codes format with their link embedded).
- Acknowledgments of funding agencies (EU & SERI).

The brochure and leaflet were distributed via the project website and social media channels, and they will be used to inform the stakeholders about the project at physical events. The brochure and leaflet were also uploaded to the project website in the "Public Documents" tab under the "Communication Materials" section and in the "News" tab.

3.5.2. Poster

The project poster (presented in Annexe 6) is designed to meet the needs of the project to present information at relevant conferences and exhibitions. Additionally, the poster was also distributed on social media for broader communication. The poster provides general information on the project (challenge, solution, innovation pillars), methodology concept, objectives, project's impact, consortium, QR codes with links on social media (LinkedIn, X), acknowledgments of EU & SERI. The poster was uploaded to the project website in the "Public Documents" tab under the "Communication Materials" section in digital and printable format and in the "News" tab.

3.6. Introductory presentation of the project

A general introductory presentation of the MaaSiveTwin project towards the general public was prepared to serve as a basis for current and future presentations. It provides a brief overview

of the project and the partners, along with their geographical location. The presentation also describes the project challenge, the proposed solution, its key and specific objectives, the innovation pillars of the project, the methodology concept and its key elements, the project's impact, and the social media page's name.

The presentation is customisable to consider the audience's specific needs. The presentation will be updated with the latest information about the project as soon as results become available. The introductory presentation is provided in Annexe 7.

3.7. Researchers' introduction

Short introductory spots of MaaSiveTwin's researchers involved in the project will be created throughout the project, introducing the lead scientists/researchers/contributors and their views about the project. During these short interviews, questions about the importance of the MaaSiveTwin project, its impact, and their specific role in the project will be answered. The researchers' spots will be posted on the project's social media accounts (LinkedIn, X), featured in newsletters and uploaded to the project website in the "News" tab.

3.8. Project videos

A maximum of a 2-minute introductory video encapsulating the essence of the MaaSiveTwin project will be prepared by the end of March 2025 (month 12). The video is an excellent way to capture the attention and raise stakeholders' awareness thanks to its short duration and attractive presentation. To make it more accessible, the voiceover will be subtitled in English. The video will be communicated on social media and showcased at future events. The video will also be placed on the website homepage.

3.9. Social media

Social media is a very effective means of communicating and disseminating information fast across heterogeneous target groups and maintaining constant engagement with target groups. To extend the project audience and involve the general public (not only sector experts), MaaSiveTwin is strategically integrating these social media tools into its dissemination and communication activities. The MaaSiveTwin project aims to have a strong presence on social media to enhance its capacity to reach out to target audiences and the general public and ensure active interaction with them. The MaaSiveTwin activities on social media seek to achieve the following impacts:

- Make the project visible and disseminate news about project activities and achievements.
- Engage people in online conversations and gain a deeper insight into their views and feelings on the topics covered.
- Ensure effective real-time presentation of participation in events.
- Support the project networking activities.

LinkedIn and X were selected as the most appropriate social networks to promote the project news, outcomes and achievements.

Talos will act as the moderator of the social media accounts, managing the content, controlling and filtering inadequate content, and monitoring the suitability and relevance of the information to be published.

3.9.1. LinkedIn

The MaaSiveTwin LinkedIn page is extensively used for networking purposes, enabling the promotion of MaaSiveTwin amongst a broad community of professionals in the fields of supply chain management of CRMs, modelling and MaaS. Furthermore, MaaSiveTwin will seek opportunities for interaction with representatives of other relevant groups, such as the battery industry, energy technologies and magnet manufacturers, producers in CRM, recyclers, academic and research groups, industrial players, and other relevant EU-funded projects.

The MaaSiveTwin LinkedIn account is: <https://www.linkedin.com/showcase/maasive-twin/>

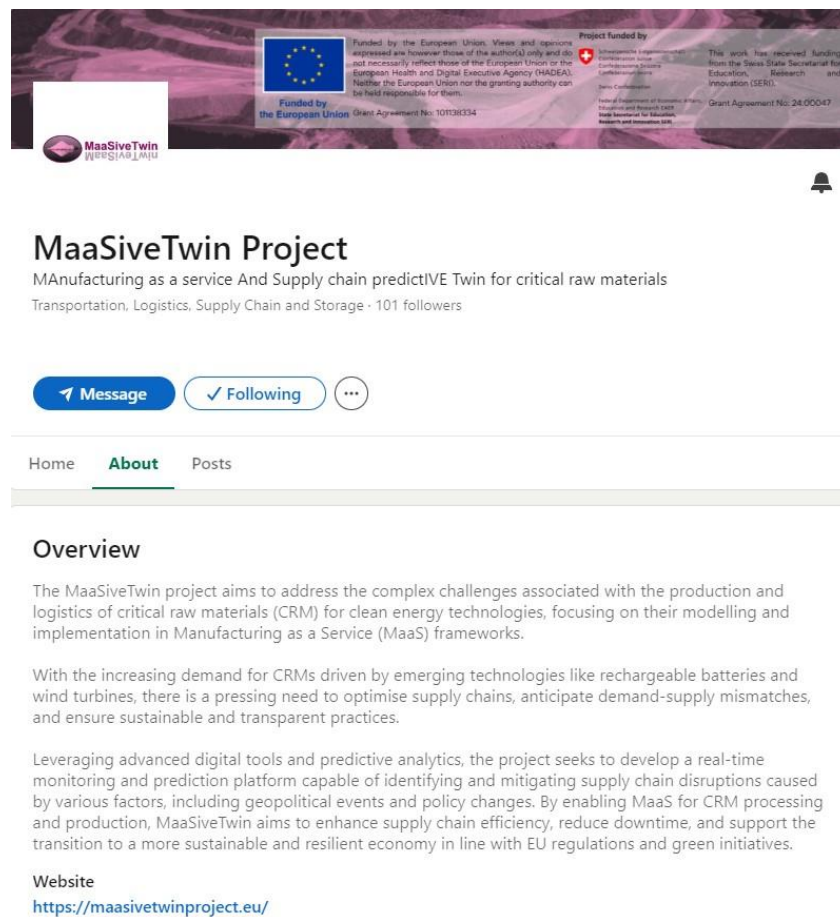


Figure 4: MaaSiveTwin LinkedIn profile

3.9.2. X (formerly Twitter)

The MaaSiveTwin X account is used for amplifying communication (both Business-to-Business and Business-to-Citizens) to a large community of active stakeholders and for spreading the news about project developments, among others. Regular posts will focus on attracting and engaging with target audiences, leading to establishing a trusted MaaSiveTwin network, enlarging the outreach to broad and targeted audiences.

The MaaSiveTwin X account is: <https://x.com/MaaSiveTwinEU>



Figure 5: MaaSiveTwin X profile

3.10. Industrial fairs/exhibitions, scientific conferences and publications

Partners will reach out to potential end users via industrial fairs and exhibitions during which dissemination workshops will be organised to present the project and to evaluate the interest towards MaaSiveTwin's new technology. Furthermore, articles will be published in specialised newsletters and industry-focused magazines/trade journals targeting specific audiences, including the newsletters of partners and stakeholder group members. Research partners will actively contribute to scientific conferences and publish papers in peer-reviewed scientific journals.

TALOS has prepared and will maintain lists (via SharePoint) with possible industry and scientific events as well as industrial magazines and scientific journals of interest to MaaSiveTwin. These lists are being updated throughout the project, and partners are able to consider and suggest new events, scientific journals or industrial magazines of interest to the project. The lists include essential event information, such as start date, end date, location, a brief description/comments and a link to the event website. The list of journals and magazines includes information such as publisher, brief description, peer-review policy, open access option and a link to the journal/magazine website (see Annexe 8).

For both lists, there is a dedicated column where partners within the consortium can declare interest in attending events or publishing in journals and magazines. This process facilitates

the constant evaluation of the dissemination and communication efforts during the project and enables the monitoring of the project's Key Performance Indicators (see related KPIs in section 4.3). Table 3 shows some of the identified events, scientific journals, and magazines.

The MaaSiveTwin consortium will also target some high-visibility events aimed at addressing the general public during the second half of the project, such as the European Researchers' Night, Science on Stage Festival, and European Science Festival. The event list below is indicative for the partners, and this list will be updated throughout the project period.

Events (Exhibitions and Conferences)		
Title	Starting Date	Link (website)
The Battery Show Europe	3-Jun-25	https://www.thebatteryshow.eu/en/home.html
Battery Experts Forum	5-Nov-24	https://www.battery-experts-forum.com/
Battery Innovation Days 2024	26-Nov-24	https://battery2030.eu/news/happenings-events/battery-innovation-days-2024/
Battery Technologies Summit	25-Sep-24	https://bepassociation.eu/battery-technologies-summit-2024/
Raw Materials Week 2024	9-Dec-24	https://single-market-economy.ec.europa.eu/sectors/raw-materials/week_en
CRM for a Sustainable Future	7-Oct-24	https://eurogeosurveys.org/event/crm-for-a-sustainable-future/
IRTC25: From Raw Material Policies to Practice	19-Feb-25	https://irtc.info/irtc-2025/
Raw Materials Summit	13-May-25	https://eitrmsummit.com/
Supply Chain Digitalization Conference & Exhibition 2025	1-May-25	https://supplychain-conference.com/
European Procurement & Supply Chain Excellence Summit 2025	31-Mar-25	https://www.bme.de/veranstaltungen/grossveranstaltungen/european-procurement-supply-chain-excellence-summit-2025/
Supply Chain Visibility Conference & Exhibition 2025	6-May-25	https://visibility-conference.com/
Data Driven Supply Chain Conference 2025	30-Apr-25	https://ptnevents.com/conferences/ddsc2025/
Sustainable Supply Chain Exhibition	25-Mar-25	https://www.sustainablesupplychainexhibition.co.uk/
IEEE International Conference on Imaging Systems & Techniques	14-Oct-25	https://ist2024.ieee-ims.org/
IEEE International Conference on Tools with Artificial Intelligence (ICTAI)	Oct-25	https://ieeexplore.ieee.org/xpl/conhome/1000763/all-proceedings
International Conference on Geographical Information Systems Theory, Applications and Management (GISTAM 2025)	1-Apr-25	https://gistam.scitevents.org/Home.aspx
Scientific Journals and Industrial Magazines		
Title	Type	Link
The Electrochemical Society	Scientific journal	https://iopscience.iop.org/journal/1944-8783

Electrochimica Acta	Scientific journal	https://www.sciencedirect.com/journal/electrochimica-acta
CHEManager	Industrial magazine	https://www.chemanager-online.com/en/chemanager-international
AutoMotorSport	Industrial magazine	https://www.auto-motor-und-sport.de/
Euractiv	Online media	https://www.euractiv.com/
Journal of Applied Physics	Scientific journal	https://pubs.aip.org/aip/jap
IEEE Transactions on Industrial Electronics	Scientific journal	https://ieeexplore.ieee.org/xpl/RecentIssue.js?punumber=41
Journal of Supply Chain Management (JSCM)	Scientific journal	https://onlinelibrary.wiley.com/journal/1745493X
Supply Chain Management	Scientific journal	https://journals.open.tudelft.nl/jscms/index
Journal of Operations Management	Scientific journal	https://onlinelibrary.wiley.com/journal/18731317
Journal of Purchasing and Supply Management	Scientific journal	https://www.sciencedirect.com/journal/journal-of-purchasing-and-supply-management
International Journal of Physical Distribution & Logistics Management	Scientific journal	https://www.emerald.com/insight/publication/issn/0960-0035
International Journal of Operations & Production Management	Scientific journal	https://www.emerald.com/insight/publication/issn/0144-3577
Production and Operations Management	Scientific journal	https://journals.sagepub.com/home/paoa
Sustainable Materials and Technologies	Scientific journal	https://www.sciencedirect.com/journal/sustainable-materials-and-technologies
Resources, Conservation and Recycling	Scientific journal	https://www.sciencedirect.com/journal/resources-conservation-and-recycling
IEEE Transactions on Artificial Intelligence	Scientific journal	https://ieeexplore.ieee.org/xpl/RecentIssue.js?punumber=9078688
WILEY Sustainable Development	Scientific journal	https://onlinelibrary.wiley.com/journal/10991719
MDPI Remote Sensing, Advances in Satellite Image Analysis and Applications for Earth Observation	Scientific journal	https://www.mdpi.com/journal/remotesensing/special_issues/SP865S162Y
Earth Imaging Journal	Scientific magazine	https://eijournal.com
Geosciences	Scientific journal	https://www.mdpi.com/journal/geosciences

Table 3: Events-Journals-Magazines of interest to MaaSiveTwin

3.11. Thematic workshops

MaaSiveTwin partners will organise a series of thematic-dedicated workshops designed to engage specific target groups. These workshops will provide a platform for direct interaction with relevant stakeholders, including representatives from umbrella organizations, thereby leveraging their multiplier effect to further extend the project's outreach. Individual invitations will be extended to ensure the active participation of those with the most influence and interest in the topics.

A core objective of these workshops is to foster bi-directional communication between MaaSiveTwin project and the broader community, allowing for an open exchange of ideas, insights, and feedback. MaaSiveTwin aims to align its goals with stakeholder expectations and ensure meaningful engagement throughout the project's lifecycle by creating an interactive and collaborative environment.

Each of the four workshops stated in the GA will focus on a distinct theme and will be hosted by different partner in their relevant field:

1. Sustainability tracking in CRM supply chains:

- **Host:** TUHH
- **Where:** Hamburg, Germany
- **When:** M36 (Spring 2027)
- **Engaged stakeholders:** scientific/research community, industrial organizations
- **Description:** This workshop will explore the topic of sustainability tracking within CRM supply chains. Selected results of the MaaSiveTwin project will be presented, complemented by invited keynotes from researchers, industry leaders, and/or policymakers. The primary goal is to engage the key stakeholders in discussions that highlight best practices, challenges, and opportunities in sustainability tracking. By gathering insights and feedback, we aim to refine our approaches and develop actionable strategies that enhance the sustainability of CRM supply chains.

2. Deterministic Modelling of Supply-Demand Mismatch:

- **Host:** BATT
- **Where:** Hamburg, Germany
- **When:** M36 (Spring 2027)
- **Engaged stakeholders:** scientific/research community, industrial organizations
- **Description:** This workshop will focus on the development and use of deterministic, physics-based models to predict and manage supply-demand mismatches within the CRM industry. Selected results of the MaaSiveTwin project will be presented, for example, forecasting price fluctuations of raw materials, assessing supply chain sustainability and risks, and/or presenting data for accurate multi-source forecasting. Participants will explore how these models can support competitive assessments of locally produced EU goods versus international alternatives, factoring in variables like raw material costs, tariffs, and production methods.

3. Data Fusion (DF) and Machine Learning (ML) Modelling:

- **Host:** TUC
- **Where:** Chania, Greece
- **When:** around M27 (Summer 2026)
- **Engaged stakeholders:** industrial organisations & scientific/research community
- **Description:** This workshop will focus on presenting basic aspects of DF and ML modelling, including the main results and pitfalls of WP4. The audience will be scientific and non-scientific, as well as key stakeholders. The primary aim is to gather valuable insights and feedback from the audience to ensure that the ML outcomes can have a meaningful translation in real-world applications. Stakeholder

input will be carefully evaluated, with actionable suggestions analysed and incorporated to refine and enhance the DF / ML procedures and relatively improved results.

4. Successful Exploitation, Communication and Dissemination:

- **Host:** TALOS
- **Where:** Cyprus
- **When:** around M42 (Autumn 2027)
- **Engaged stakeholders:** industrial organisations & scientific/research community
- **Description:** This workshop will focus on presenting both the scientific and non-scientific results of the project to key stakeholders. The primary aim is to gather valuable insights and feedback from these stakeholders to ensure that the outcomes can be effectively translated and applied in real-world settings. Stakeholder input will be carefully evaluated, with actionable suggestions analysed and incorporated to refine and enhance the project's results. Stakeholders' engagement in this workshop will help ensure that the project's outcomes are not only disseminated but also optimized for maximum impact and practical application.

4. Dissemination reporting, monitoring and evaluation

Talos, as leader of WP6 – Communication, dissemination and exploitation, is responsible for coordinating communication and dissemination activities. Nevertheless, all project partners are obligated to contribute to the implementation of relevant actions. Partners are responsible for providing necessary information about their work and the results achieved and for actively participating in the planning, execution, and reporting of communication and dissemination activities.

4.1. Monitoring and reporting of performed activities

To ensure the successful implementation of the communication and dissemination activities, systematic monitoring will be carried out throughout the project. Regular monitoring will allow for the identification of possible risks and deviations from the Dissemination and Communication Plan and relevant key performance indicators and the timely planning of any necessary corrective actions to address potential implementation problems. Such an approach will improve the overall performance of the relevant activities and enable a more efficient evaluation.

The monitoring will be performed:

- A. *Internally on a regular basis:* three online forms have been created via SharePoint for reporting all Dissemination, Communication and Publication activities by the partners. The forms are available at Microsoft SharePoint and accessible to all partners. Partners must fill out these forms whenever they carry out a dissemination and communication activity or report the activity to the WP6 leader TALOS.
- B. *Externally through submission of:*

- a. Periodic reporting (via SyGMA platform): during the periodic reporting, the relevant data regarding performed Dissemination, Communication and Publication activities will be filled in on the project reporting platform, and WP6 activities will be reported in the Technical Reports.
- b. Deliverables:
 - i. D6.1 – Communication, dissemination, engagement strategic plan: Report outlining the strategic plan of (1) approaches (2) detailed actions (3) KPIs and methodology for evaluation of communication, dissemination and engagement activities
 - ii. D6.2 – Outcomes from the clustering and cooperation activities: Reports on clustering and networking to promote digitalization and transparency in the CRM value chain.
 - iii. D6.3 – MaaSiveTwin market analysis & key exploitable results: Reports on the exploitation plan for all the Key Exploitable Results which details how each of the project outputs will be used in non-commercial & commercial exploitation activities
 - iv. D6.4 – Innovative business models and IPR strategy: Report outlining the strategy to protect the results through different methods depending on the nature of the results: patent, copyright, trademark, contractual, etc. Applying MaaSiveTwin demonstrator to do market analysis, a comprehensive business model for different battery material production routes will be analysed.

4.2. Evaluation of performed activities

In order to assess the impact of a dissemination and communication activity, the following must be evaluated:

- Quantity: volume – e.g. the quantity of people reached via the dissemination activity
- Quality: effectiveness – e.g. how convincing the dissemination action was

As for the volume, MaaSiveTwin project partners will estimate the number of people reached during the dissemination action.

Regarding the effectiveness, a specific ranking will be used, as shown in the table below (specific KPIs are given in Section 4.3):

RATING	PRESENTATION	PUBLICATION	WEBSITE
LOW	Marginal interest, no questions	No references	No site visits, no external links
MEDIUM	Good interest, many questions, some new contacts	Few references	Few site visits, limited external links
HIGH	High interest, many questions and appreciations, contacts and following actions	Many references	Many site visits and external links

Table 4: Effectiveness ranking for dissemination and communication activities depending on the tool used

Based on this classification, the impact of each dissemination and communication activity will be assessed and benchmarked towards the pre-defined KPIs. Accordingly, specific actions will be suggested by the WP6 leader (TALOS), and the current Dissemination and Communication Plan will be updated if necessary.

4.3. Key performance indicators

MaaSiveTwin partners have identified a series of Key Performance Indicators (KPIs) that have been set as targets for dissemination and communication activities. As the current deliverable is a living document, these KPIs will be adjusted based on the monitoring and evaluation of the dissemination and communication actions performed or any other unforeseen events. As first targets, the following KPIs have been defined by the partners:

Dissemination channels and activities	KPIs
Scientific publications in peer-reviewed journals	>3
Technical magazines	>5 articles
Participation in conferences & fairs	>20 events
Thematic workshops	4 workshops (30+ participants each)
MaaSiveTwin main outreach final event	>60 participants
Participation in clustering with related HEU/H2020 & other initiatives	>2 joint activity/meetings
Meetings with regional & EU stakeholders	>3 meetings at events
Communication channels and activities	KPIs
MaaSiveTwin branding (project visual identity)	1 brand
Project brochure	>2000 distributed
Project leaflet	>2000 distributed
Project poster	1
Online/printed press releases	>4; in ENG & translated into 3 other languages
Public electronic newsletters	9; >100 subscriptions
Project videos	2; >1000 viewers
Project website	>100 visits/month
Project LinkedIn account	>300 followers
Project X account	>300 followers

Table 5: Key Performance Indicators for dissemination and communication activities

5. Networking

MaaSiveTwin project will not limit its dissemination and communication activities to the goals of awareness and understanding. More than informing, the project will aim to trigger involvement in and endorse its activities and results, seeking to enhance links and synergies with similar initiatives. In order to enhance the visibility of the MaaSiveTwin project and to ensure maximum attractiveness and engagement of the stakeholders, a series of liaison and networking activities will take place within the project lifespan. This activity will have both a scientific and a business focus, as well as seek collaboration and establish a dialogue with

different EU-funded H2020 and Horizon Europe projects and initiatives. This will be an ongoing activity that will enable the exchange of views and the potential development of synergies.

The main joint activities targeted with other relevant projects might include the exchange of best practices, exchange of “lessons learnt”, organisation of joint dissemination and communication events, facilitating social media presence (sharing, following, re-tweeting, etc.), joint educational activities, joint research activities, shared use of research infrastructure as well as the preparation of jointly performed market/industry surveys. An initial list of potential projects for collaboration is provided in Table 6 below. This table will be updated with future relevant projects funded under Horizon Europe. MaaSiveTwin consortium partners will select the most relevant projects for networking activities, indicating the possible synergies and topics for discussion. The coordinator and the dissemination and communication manager of the MaaSiveTwin project will contact the coordinators of the selected projects with the aim of discussing with them potential collaboration either from a scientific and industrial perspective or with the aim of organising joint dissemination and communication activities. The first contact is scheduled to take place around month 8 of the project, followed up by thematic workshops later on.

Acronym	Title	Cordis link
M4ESTRO	Industrial Manufacturing As a sERvice STRategies and models for flexible, resilient, and reconfigurable value networks through Trusted and Transparent Operations.	https://cordis.europa.eu/project/id/101138506
ACCURATE	ACHIEVING RESILIENCE THROUGH MANUFACTURING AS A SERVICE, DIGITAL TWINS AND ECOSYSTEMS	https://cordis.europa.eu/project/id/101138269
Tec4MaaSEs	Technologies for Manufacturing as a Service Ecosystems	https://cordis.europa.eu/project/id/101138517
MAASive	MANUFACTURING AS A SERVICE TO INCREASE RESILIENCE IN VALUE NETWORKS	https://cordis.europa.eu/project/id/101138040
NARRATE	regeNerAtive Resilient smaRt mAnufacTuring nEtworks	https://cordis.europa.eu/project/id/101138094
s-X-AIPI	self-X Artificial Intelligence for European Process Industry digital transformation	https://cordis.europa.eu/project/id/101058715
AIDEAS	AI Driven industrial Equipment product life	https://cordis.europa.eu/project/id/101057294

	cycle boosting Agility, Sustainability and resilience	
Circular TwAIIn	AI Platform for Integrated Sustainable and Circular Manufacturing	https://cordis.europa.eu/project/id/101058585
H4C ECoP	European Community of Practice supporting Hubs for Circularity (H4C ECoP)	https://cordis.europa.eu/project/id/101058656
H4C EUROPE	Building a European Community of Practice of Hubs for Circularity	https://cordis.europa.eu/project/id/101058416
Fluently	Fluently - the essence of human-robot interaction	https://cordis.europa.eu/project/id/101058680
CONVERGING	Social industrial collaborative environments integrating AI, Big Data and Robotics for smart manufacturing	https://cordis.europa.eu/project/id/101058521
AI-PRISM	AI Powered human-centred Robot Interactions for Smart Manufacturing	https://cordis.europa.eu/project/id/101058589
COGNIMAN	COGNitive Industries for smart MANufacturing (COGNIMAN)	https://cordis.europa.eu/project/id/101058477
ZDZW	Non-Destructive Inspection Services for Digitally Enhanced Zero Waste Manufacturing	https://cordis.europa.eu/project/id/101057404
OPENZDM	OPEN PLATFORM FOR REALIZING ZERO DEFECTS IN CYBER PHYSICAL MANUFACTURING	https://cordis.europa.eu/project/id/101058673
MASTT2040	MANUFACTURING AS A SERVICE FOR THE EU'S TWIN TRANSITION UNTIL 2040	https://cordis.europa.eu/project/id/101138342
RE-EURECA-PRO	REsearch and innovation dimension of the European University on REsponsible Consumption And PROduction	https://cordis.europa.eu/project/id/101035798
Plooto	Product Passport Through Twinning of Circular Value Chains	https://cordis.europa.eu/project/id/101092008

DECISO	DEvelopers of Circular SOLUTIONS	https://cordis.europa.eu/project/id/101082232
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Table 6: List of relevant projects

6. Appendices

Annexe 1 – PowerPoint Presentation template



Figure 6: MaaSiveTwin presentation template - Title slide



Figure 7: MaaSiveTwin presentation template - Main slide



MaaSiveTwin

niwTevISaam

MANUFACTURING AS A SERVICE AND SUPPLY CHAIN PREDICTIVE TWIN FOR CRITICAL RAW MATERIALS








THANK YOU FOR YOUR ATTENTION!

 @MaaSiveTwinEU
 @MaaSiveTwin project



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Figure 8: MaaSiveTwin presentation template - Closing slide

Annexe 2 – Project website images



The screenshot displays the MaaSiveTwin project website homepage, which is designed with a dark purple and black color scheme. The main header features the project title "MANUFACTURING AS A SERVICE AND SUPPLY CHAIN PREDICTIVE TWIN FOR CRITICAL RAW MATERIALS" in white and pink text, accompanied by a 3D rendering of a red industrial ship. Below this, the "CHALLENGE" section describes the complexities of managing CRM supply chains, supported by an image of a white "LITHIUM-ION ENERGY STORAGE" unit. The "SOLUTION" section outlines the goal of developing a platform for early decision-making, illustrated with a laptop showing data visualizations. A "PROJECT DETAILS" button is visible between the challenge and solution sections.

IMPACT

The MaaSiveTwin project is poised to significantly impact various dimensions, addressing key challenges in the CRM value chain and contributing to a more efficient, resilient, and sustainable industrial ecosystem.

Supply Chain Optimization

MaaSiveTwin aims to optimise the CRM value chain, enabling companies to make informed decisions based on comprehensive data and analysis. This includes adopting production schedules, mitigating supply chain disruptions, and identifying alternative supply sources, ultimately leading to reduced logistic costs, increased utilization rates, and reduced variable costs.

Resilience and Sustainability

MaaSiveTwin enhances the resilience and sustainability of industrial systems and value chains. It provides insights into the expected CRM demand, enabling swift adaptation to external conditions, and facilitates the identification of mitigation strategies in case of threats. Additionally, it promotes sustainable management practices and improves policy-making related to environmental protection and sustainability, thereby fostering a more resilient and sustainable industrial ecosystem.

Global Leadership and Collaboration

MaaSiveTwin contributes to EU digitalisation efforts by fostering collaboration across various cultures and expertise domains. It aims to upskill human capital in the IT sector, promote multidisciplinary collaboration for generating deep data, and increase global public awareness of sustainable management practices.

MAASIVETWIN FAST FACTS

Horizon-RIA	Starting Day	Months Duration
MaaSiveTwin is a fully funded Horizon Europe - Research & Innovation Action	April 1st 2024	48

CONSORTIUM

Logos of consortium members: Balyonics (DE), ElevateS, RockTech, Balyonics (PL), TUHH, and others.

SUBSCRIBE TO OUR NEWSLETTER

Form fields for Name and Email, with a SUBSCRIBE button.

PROJECT COORDINATOR

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DISSEMINATION & COMMUNICATION

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Contract No. 24.00047

Figure 9: MaaSiveTwin website – Homepage

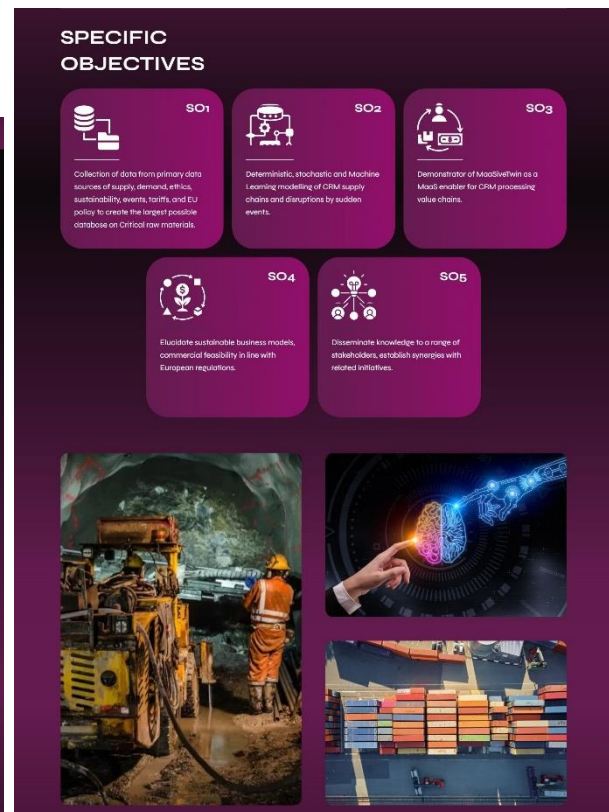
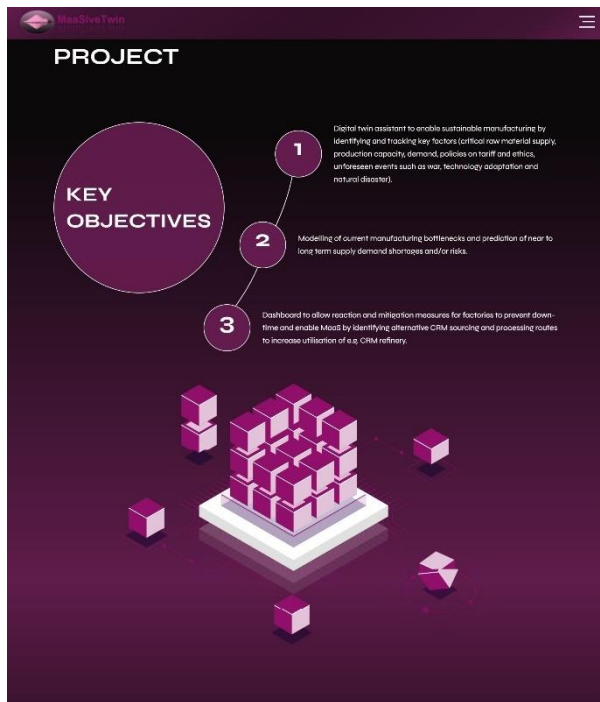


Figure 10: MaaSiveTwin website – Project tab; Key Objectives & Specific Objectives

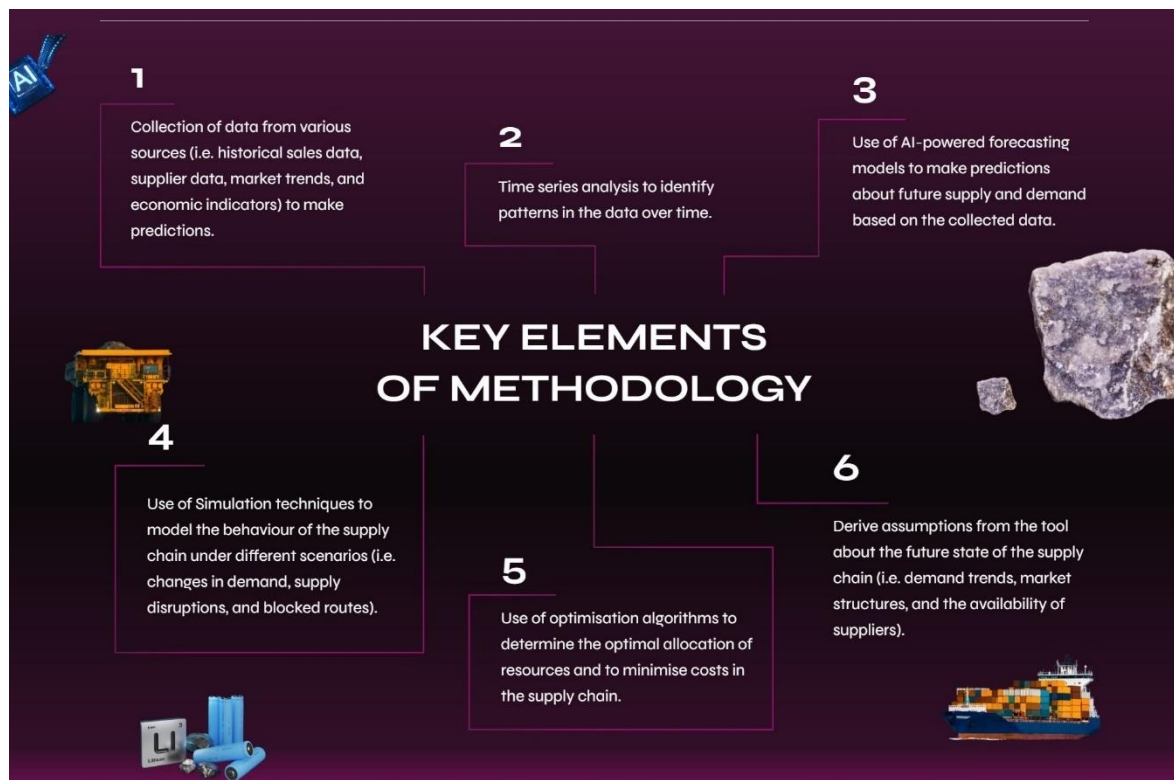


Figure 11: MaaSiveTwin website – Project tab; Key elements of Methodology

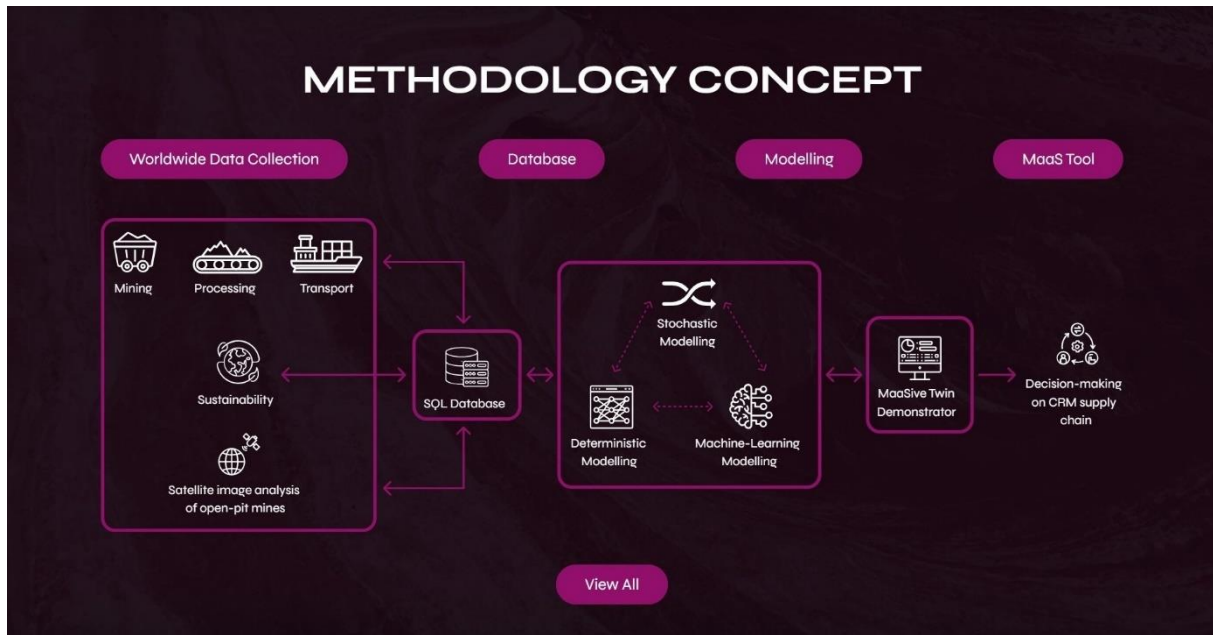
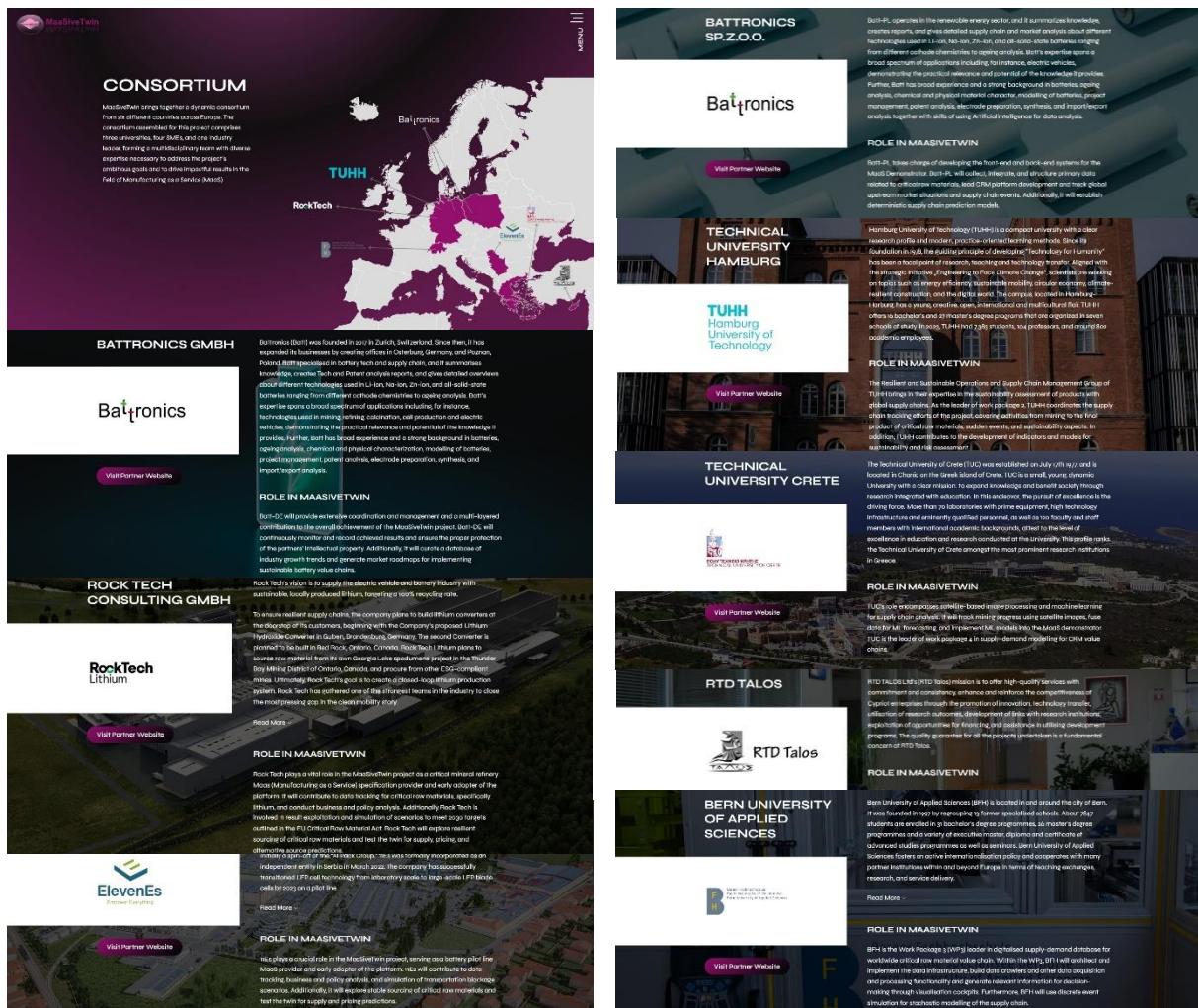


Figure 12: MaaSiveTwin website – Project tab; Methodology concept (interactive element)



The screenshot displays the Consortium tab of the MaaSiveTwin website, featuring a grid of consortium members and their roles in the project.

CONSORTIUM

MaaSiveTwin brings together a dynamic consortium from 16 different countries across Europe. The consortium assembled for this project comprises three universities, four SMEs, and one industry leader, forming a multidisciplinary team with diverse expertise necessary to address the project's overall goals and to develop impactful results in the field of Manufacturing as a Service (MaS).

BATTRONICS GMBH

Batronics GmbH was founded in 1991 in Datteln, Switzerland. Since then, it has expanded its business by opening offices in Datteln, Germany, and Pöchlarn, Austria. Batronics specializes in battery tech and supply chain, and it summarizes knowledge, creates both an internal expertise report, and does external work on various different technologies used in Li-Ion, Na-Ion, Zn-Ion, and solid-state batteries ranging from different cathode chemistries to ageing analysis. Bat's expertise spans a broad spectrum of applications including, for instance, technologies used in mining, farming, construction, and industries and electric vehicles, demonstrating the practical relevance and potential of the knowledge it provides. Bat has, for example, a strong background in battery management, ageing analysis, electrode preparation, synthesis, and import/export analysis.

ROCK TECH CONSULTING GMBH

Rock Tech's vision is to supply the electric vehicle and battery industry with sustainable, locally produced lithium, lowering the total recycling rate.

ElevenEs

ElevenEs is a leading provider of digital supply chain solutions, specializing in the automotive and industrial sectors. The company has successfully implemented digital supply chain solutions for various clients, including the automotive and industrial sectors.

BATTRONICS SP. Z.O.O.

Batronics operates in the renewable energy sector, and it summarizes knowledge, creates reports, and gives detailed supply chain and market analysis about different technologies used in Li-Ion, Na-Ion, Zn-Ion, and solid-state batteries ranging from different cathode chemistries to ageing analysis. Bat's expertise spans a broad spectrum of applications including, for instance, electric vehicles, demonstrating the practical relevance and potential of the knowledge it provides. Bat has, for example, a strong background in battery management, ageing analysis, electrode preparation, synthesis, and import/export analysis.

TECHNICAL UNIVERSITY HAMBURG

Hamburg University of Technology (TUHH) is a compact university with a clear research profile and modern, practice-oriented learning methods. Since its foundation in 1978, the guiding principle of developing "Technology for Humanity" has been a central point of research, teaching and technology transfer. Integral to the strategic initiative "Engineering to face climate change" are solutions aiming on topics such as energy efficiency, sustainable mobility, circular economy, climate-resilient engineering, and the digital world. The corporate research and innovation unit has a broad, cross-disciplinary, interdisciplinary and multidisciplinary TUHH offers a bachelor's and a master's degree programs that are organized in seven schools of study. Among TUHH's most study programs, its professors, and research for sustainable development.

TECHNICAL UNIVERSITY CRETE

The Technical University of Crete (TUC) was established on July 1971, and is located in Chania on the Greek island of Crete. TUC is a small, young, dynamic University with a clear vision: to expand knowledge and benefit society through research, teaching and education. In this vision, the pursuit of excellence is the driving force. More than 30 laboratories with prime equipment, high technology infrastructure and scientifically qualified personnel, as well as two faculty and staff members with a professional academic background, offer to the field of excellence in education and research conducted at the university. This profile makes the Technical University of Crete among the most prominent research institutions in Greece.

RTD TALOS

RTD TALOS Ltd's RTD Talos mission is to offer high-quality services with commitment and consistency, enhance and reinforce the competitiveness of Greek companies through the promotion of innovation, technology transfer, implementation of research outcomes, development of links with research institutions, implementation of opportunities for financing, and assistance in offering development programs. The quality guarantee for all the projects undertaken is a fundamental concern of RTD Talos.

BERN UNIVERSITY OF APPLIED SCIENCES

Bern University of Applied Sciences (BFH) is a leading university in the field of applied sciences, offering a wide range of bachelor's and master's degree programs. The university is committed to research, teaching, and innovation, and it has a strong focus on the development of sustainable and socially responsible solutions.

Figure 13: MaaSiveTwin website – Consortium tab

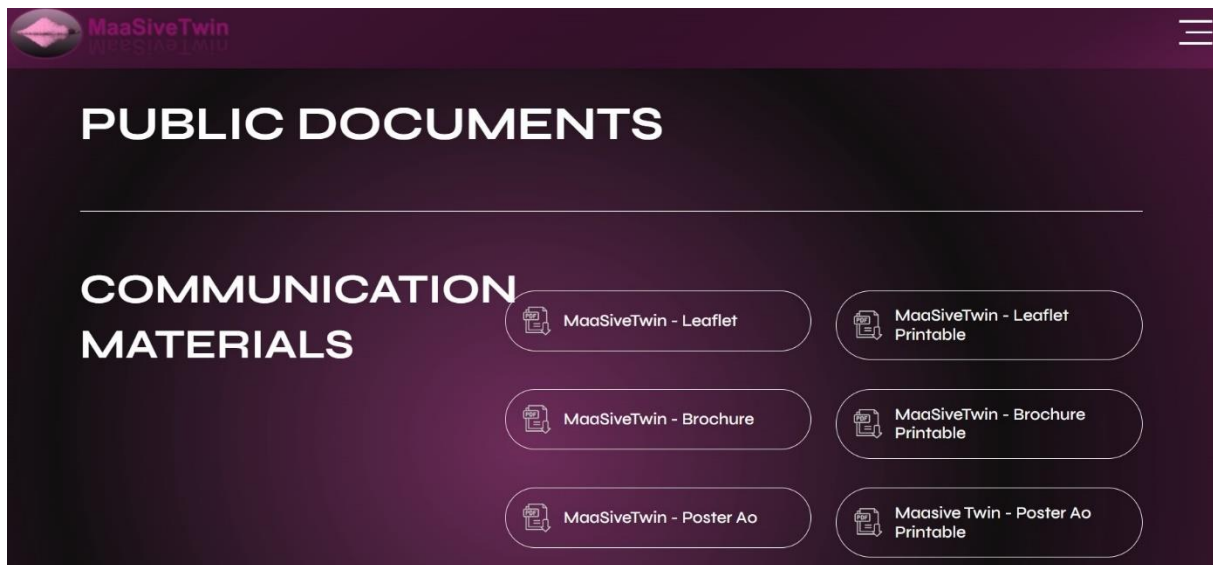


Figure 14: MaaSiveTwin website – Public Documents tab

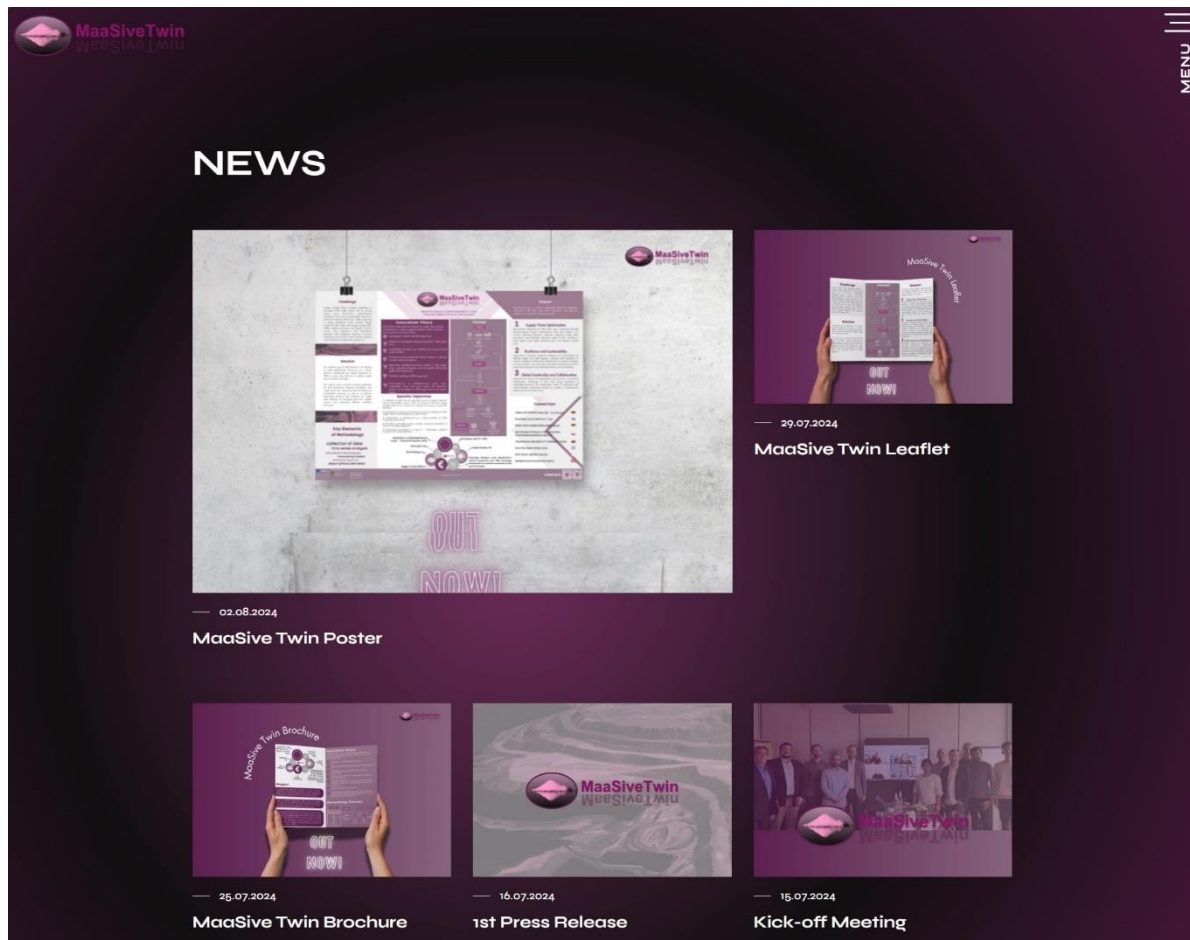


Figure 15: MaaSiveTwin website – News tab

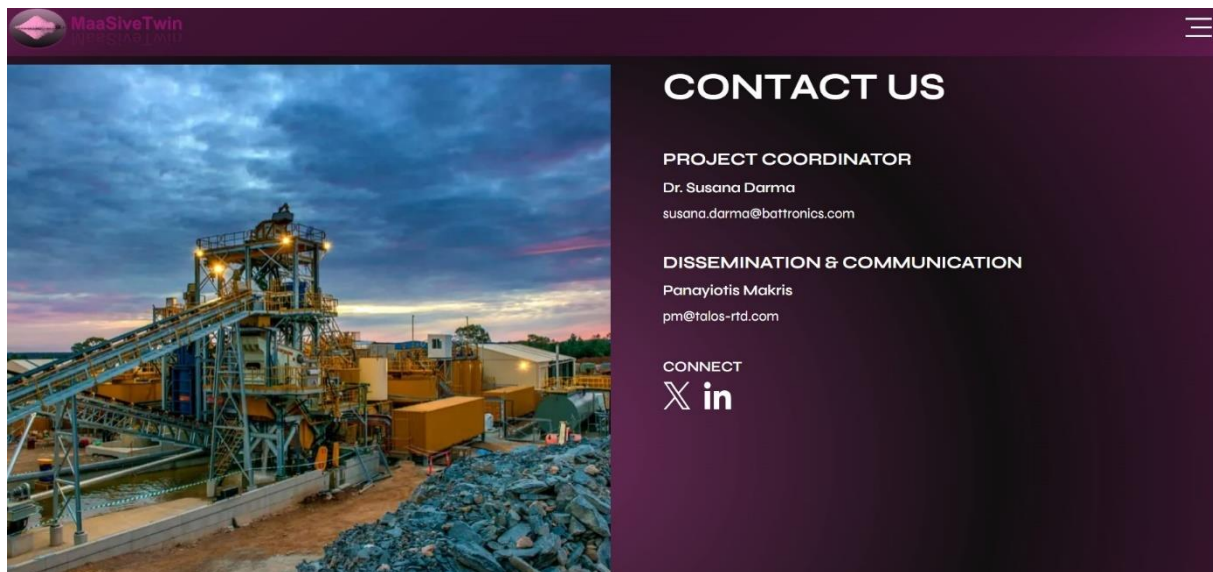


Figure 16: MaaSiveTwin website – Contact Us tab

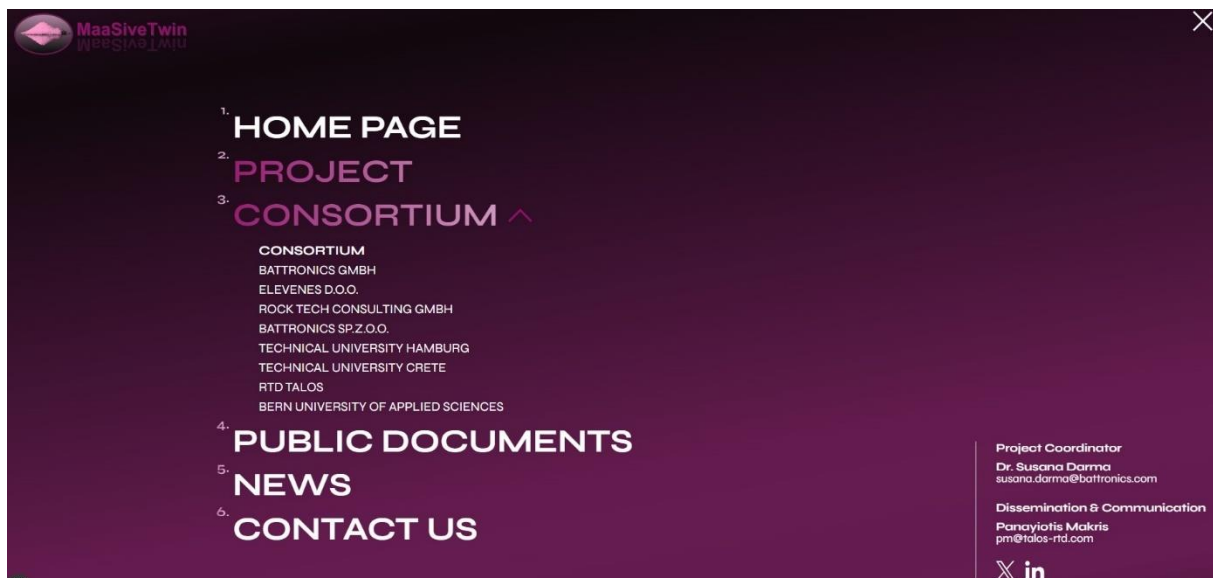


Figure 17: MaaSiveTwin website – Main Menu

Annexe 3 – 1st Press Release



MaaSiveTwin
niwTev!Saam

MaaSiveTwin – Manufacturing as a service And Supply chain predictive Twin for critical raw materials

PRESS RELEASE

Project Data

Grant Agreement ID	101138334
Starting Date	1 April 2024
Duration	48 months
Total Budget	5.2 million EUR
X Account	@MaaSiveTwinEU
LinkedIn	MaaSiveTwin Project
Cordis Page	https://cordis.europa.eu/project/id/101138334

The MaaSiveTwin project aims to address the complex challenges associated with the production and logistics of critical raw materials (CRM) for clean energy technologies, focusing on their modelling and implementation in Manufacturing as a Service (MaaS) frameworks. With the increasing demand for CRMs driven by emerging technologies like rechargeable batteries and wind turbines, there is a pressing need to optimise supply chains, anticipate demand-supply mismatches, and ensure sustainable and transparent practices. Leveraging advanced digital tools and predictive analytics, the project seeks to develop a real-time monitoring and prediction platform capable of identifying and mitigating supply chain disruptions caused by various factors, including geopolitical events and policy changes. By enabling MaaS for CRM processing and production, MaaSiveTwin aims to enhance supply chain efficiency, reduce downtime, and support the transition to a more sustainable and resilient economy in line with EU regulations and green initiatives.

Objectives

The MaaSiveTwin project contains 3 key objectives:

- Digital twin assistant to enable sustainable manufacturing by identifying and tracking key factors (critical raw material supply, production capacity, demand, policies on tariff and ethics, unforeseen events such as war, technology adaptation, and natural disaster)
- Modelling of current manufacturing bottlenecks and prediction of near to long-term supply-demand shortages and/or risks
- Dashboard to allow reaction and mitigation measures for factories to prevent downtime and enable MaaS by identifying alternative CRM sourcing and processing routes to increase utilisation of e.g. CRM refinery.

Impact

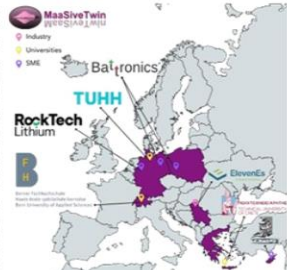
The MaaSiveTwin project is poised to have a significant impact across various dimensions:

- Supply Chain Optimization:** MaaSiveTwin aims to optimise the CRM value chain (e.g. U, Ni, Co, Pt, rare earth, etc.), enabling companies to make informed decisions based on comprehensive data and analysis. This includes adapting production schedules, mitigating supply chain disruptions, and identifying alternative supply sources, ultimately leading to reduced logistic costs, increased utilization rates, and reduced variable costs.
- Resilience and Sustainability:** MaaSiveTwin enhances the resilience and sustainability of industrial systems and value chains. It provides insights into the expected CRM demand, enabling swift adaptation to external conditions, and facilitates the identification of mitigation strategies in case of threats. Additionally, it promotes sustainable management practices and improves policy-making related to environmental protection and sustainability, thereby fostering a more resilient and sustainable industrial ecosystem.
- Global Leadership and Collaboration:** MaaSiveTwin contributes to EU digitalisation efforts by fostering collaboration across various cultures and expertise domains. It aims to upskill human capital in the IT sector, promote multidisciplinary collaboration for generating deep data, and increase global public awareness of sustainable management practices. By promoting collaboration and knowledge sharing, the project accelerates the development of IT proficiency and critical thinking skills, thereby positioning Europe as a leader in human-centric digital systems and infrastructure.

Overall, MaaSiveTwin's impact spans supply chain optimization, resilience, sustainability, global leadership, and collaboration, addressing key challenges in the CRM value chain and contributing to a more efficient, resilient, and sustainable industrial ecosystem.

Consortium

MaaSiveTwin brings together a dynamic consortium from six different countries across Europe. The consortium assembled for this project comprises three universities, four SMEs, and one industry leader, forming a multidisciplinary team with diverse expertise necessary to address the project's ambitious goals and to drive impactful results in the field of MaaS. This includes Europe's pioneers in large-scale LFP battery production (11ES) and Europe's first refinery company (Rock), a company specialized in battery tech and supply chain (Batt-DE), alongside experts in IT, modelling, forecasting, and image processing



(Batt-PL, BFH, TUC), sustainability and ethics (TUHH), and effective communication and dissemination (Talos). Together, these partners represent a collaborative effort toward achieving the project's goal of revolutionising supply chain management for critical raw materials in clean energy technologies.

PARTICIPATING ORGANISATIONS	ROLE	SHORT NAME	COUNTRY
1 FIRMA BATTRONICS	Coordinator	Batt-DE	Germany
2 ELEVENES DOO SUBOTICA	Partner	11Es	Serbia
3 ROCKTECH CONSULTING GMBH	Partner	Rock	Germany
4 BATTRONICS SPOLKA Z OGRANICZONA ODPOWIEDZIALNOSCIA	Partner	Batt-PL	Poland
5 TECHNISCHE UNIVERSITAT HAMBURG	Partner	TUHH	Germany
6 POYTECHNEO KRITIS	Partner	TUC	Greece
7 TALOS LIMITED	Partner	Talos	Cyprus
8 BERNER FACHHOCHSCHULE	Associated	BFH	Switzerland

 **Funded by the European Union**

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Health and Digital Executive Agency (HDEA). Neither the European Union nor the granting authority can be held responsible for them.

Grant Agreement No. 101138334

Project funded by

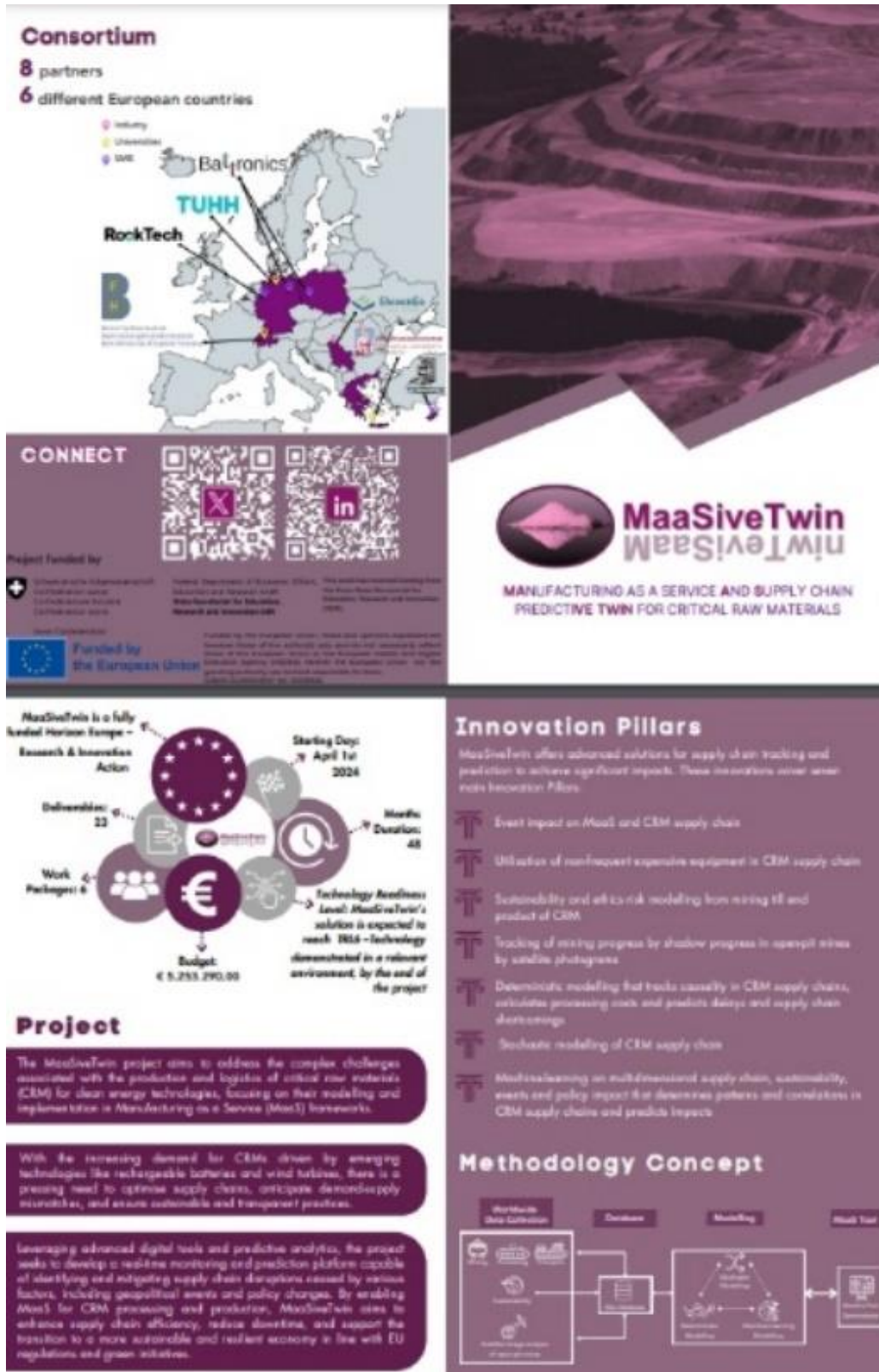

 Schweizerische Eidgenossenschaft
 Confédération suisse
 Confederaziun Svizra
 Confederaziun svizra

Federal Department of Economic Affairs,
 Education and Research (SEER)
 State Secretariat for Education,
 Research and Innovation (SERI)

Swiss Confederation
This work has received funding from the Swiss State Secretariat for Education, Research and Innovation (SERI).

Figure 18: Press Release in ENG

Annexe 4 – Brochure



Consortium
 8 partners
 6 different European countries

Industry
 Universities
 SME

Baltronics
 TUHH
 RockTech
 B
 Dusseldorf

CONNECT

Project Funded by

European Union
 German Government
 Federal Department of Economic Affairs, Research and Innovation (BMWi)

Funded by the European Union

MaaSiveTwin
 niwTev!Saam

MANUFACTURING AS A SERVICE AND SUPPLY CHAIN
 PREDICTIVE TWIN FOR CRITICAL RAW MATERIALS

MaaSiveTwin is a fully funded Horizon Europe – Research & Innovation Action

Starting Day: April 1st 2024

Duration: 48 Months

Deliverables: 23

Work Packages: 6

Budget: € 5.253.290,00

Technology Readiness Level: MaaSiveTwin's solution is expected to reach TRL8 – technology demonstrated in a relevant environment by the end of the project

Project

The MaaSiveTwin project aims to address the complex challenges associated with the production and logistics of critical raw materials (CRM) for clean energy technologies, focusing on their modelling and implementation in Manufacturing as a Service (MaaS) frameworks.

With the increasing demand for CRMs driven by emerging technologies like rechargeable batteries and wind turbines, there is a pressing need to optimise supply chains, anticipate demand/supply imbalances, and ensure sustainable and transparent practices.

Leveraging advanced digital tools and predictive analytics, the project seeks to develop a real-time monitoring and prediction platform capable of identifying and mitigating supply chain disruptions caused by various factors, including geopolitical events and policy changes. By enabling MaaS for CRM processing and production, MaaSiveTwin aims to enhance supply chain efficiency, reduce downtime, and support the transition to a more sustainable and resilient economy in line with EU regulations and green initiatives.

Innovation Pillars

MaaSiveTwin offers advanced solutions for supply chain tracking and prediction to achieve significant impacts. These innovations deliver under main Innovation Pillars:

- Event Impact on MaaS and CRM supply chain
- Utilisation of non-frequent expensive equipment in CRM supply chain
- Sustainability and ethics risk modelling from mining till end product of CRM
- Tracking of mining progress by shadow progress in openpit mines by satellite photographs
- Deterministic modelling that tracks causality in CRM supply chains, calculates processing costs and predicts delays and supply chain shortcomings
- Stochastic modelling of CRM supply chain
- MaaSiveTwin as a multidimensional supply chain, sustainability, events and policy impact that determines patterns and correlations in CRM supply chains and predicts impacts

Methodology Concept

Workflow: Data Collection → Database → Modelling → Multi Tool

Database: Data Collection, Data Storage, Data Processing, Data Analysis, Data Visualization

Modelling: Data Collection, Data Storage, Data Processing, Data Analysis, Data Visualization

Multi Tool: Data Collection, Data Storage, Data Processing, Data Analysis, Data Visualization

Figure 19: Brochure

Annexe 5 – Leaflet

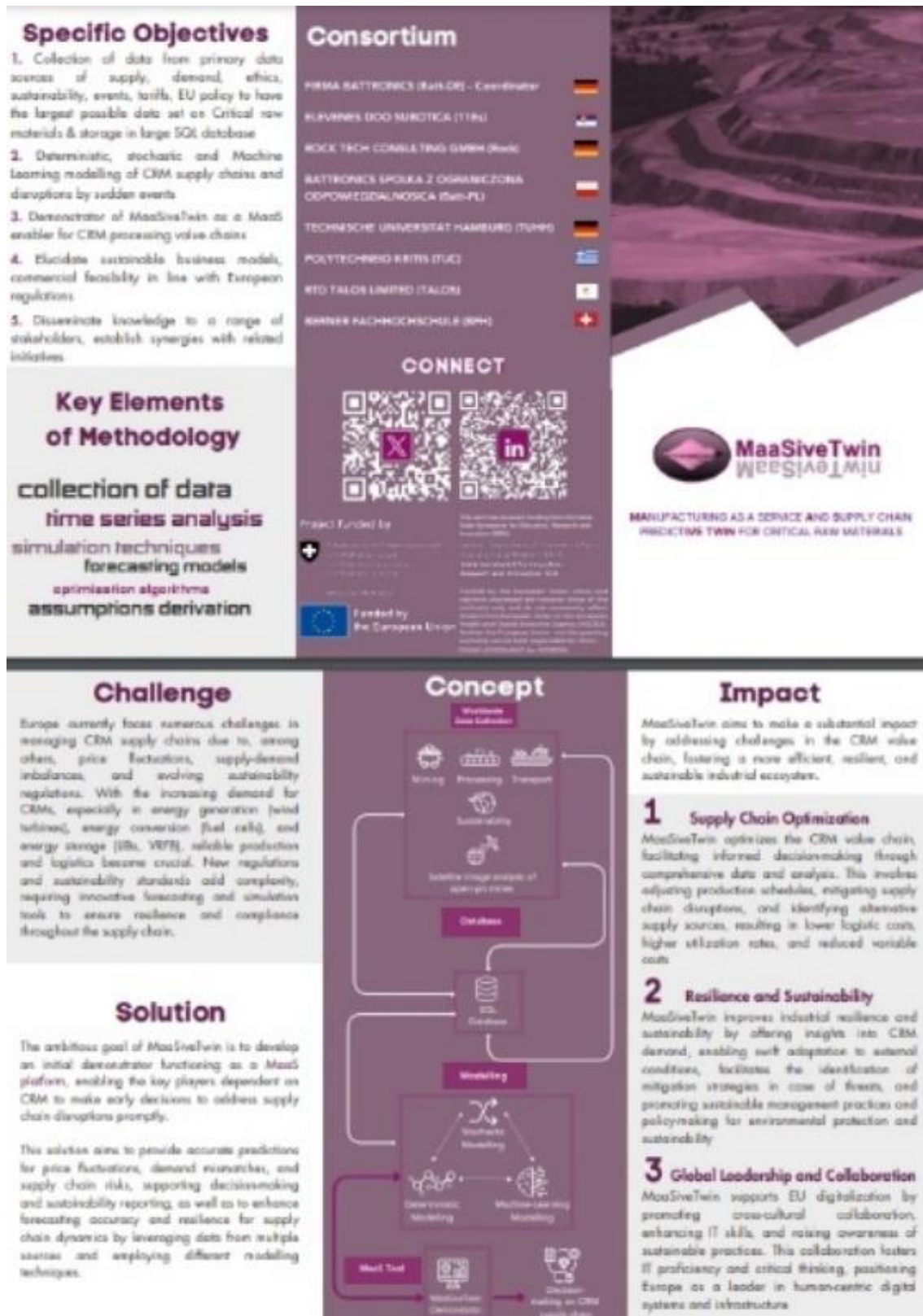
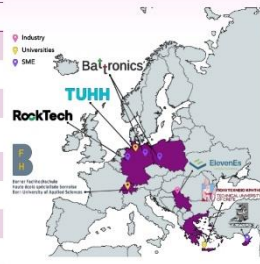


Figure 20: Leaflet

Consortium

Partner Name	Short Name	Country
1. FIRMA BATRONICS (Coordinator)	Batb DE	Germany
2. ELEVENES DOO S.R.O.	11Ex	Serbia
3. ROCKTECH CONSULTING GMBH	Rock	Germany
4. BATRONICS SPOLKA Z OGRANICZONA ODPOWIEDALNOSCIA	Batb-PL	Poland
5. TECHNISCHE UNIVERSITÄT HAMBURG	TUHH	Germany
6. POLYTECHNEO KRITIS	TUC	Greece
7. RTO TALOS LIMITED	Talos	Cyprus
8. BERNER FACHHOCHSCHULE	BFH	Switzerland



The Challenge

Europe currently faces numerous challenges in managing CRM supply chains due to, among others, price fluctuations, supply-demand imbalances, and evolving sustainability regulations. With the increasing demand for CRMs, especially in clean energy technologies (LIBs, VRFB, wind turbines, fuel cells, etc.), reliable production and logistics have become crucial. New regulations and sustainability standards add complexity, requiring innovative forecasting and simulation tools to ensure resilience and compliance throughout the supply chain.



The Solution



The ambitious goal of **MaaSiveTwin** is to develop an initial demonstrator functioning as a **MaaS platform**, enabling the key players dependent on CRM to make early decisions to address supply chain disruptions promptly.

This solution aims to provide accurate predictions for price fluctuations, demand mismatches, and supply chain risks, supporting decision-making and sustainability reporting, as well as to enhance forecasting accuracy and resilience for supply chain dynamics by leveraging data from multiple sources and employing different modelling techniques.

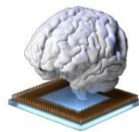
Innovation Pillars

MaaSiveTwin offers advanced solutions for supply chain tracking and prediction to achieve significant impacts. These innovations cover seven main innovation Pillars:

1. Event impact on MaaS and CRM supply chain.
2. Utilisation of non-frequent expensive equipment in CRM supply chain.
3. Sustainability and ethics risk modelling from mining till end product of CRM.
4. Tracking of mining progress by shadow progress in open-pit mines by satellite photographs.
5. Deterministic modelling that tracks causality in CRM supply chains, calculates processing costs and predicts delays and supply chain shortcomings.
6. Stochastic modelling of CRM supply chain.
7. Machine-learning on multidimensional supply chain, sustainability, events and policy impact that determines patterns and correlations in CRM supply chains and predicts impacts.

Key Objectives

1. Digital twin assistant to enable sustainable manufacturing by identifying and tracking key factors (critical raw material supply, production capacity, demand, policies on tariff and ethics, unforeseen events such as war, technology adaptation and natural disaster).
2. Modelling of current manufacturing bottlenecks and prediction of near to long term supply demand shortages and/or risks.
3. Dashboard to allow reaction and mitigation measures for factories to prevent down-time and enable MaaS by identifying alternative CRM sourcing and processing routes to increase utilisation of e.g. CRM refinery.



Specific Objectives

1. Collection of data from primary data sources of supply, demand, ethics, sustainability, events, tariffs, EU policy to have the largest possible data set on Critical raw materials & storage in large SQL database.
2. Deterministic, stochastic and Machine Learning modelling of CRM supply chains and disruptions by sudden events.
3. Demonstrator of MaaSiveTwin as a MaaS enabler for CRM processing value chains.
4. Elucidate sustainable business models, commercial feasibility in line with European regulations.
5. Disseminate knowledge to a range of stakeholders, establish synergies with related initiatives.

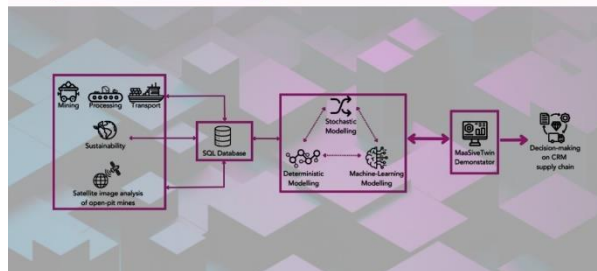


Key Elements of Methodology

Key Elements of Methodology
 collection of data
 time series analysis
 simulation techniques
 forecasting models
 optimisation algorithms
 assumptions derivation

1. Collection of data from various sources (i.e. historical sales data, supplier data, market trends, and economic indicators) to make predictions.
2. Time series analysis to identify patterns in the data over time.
3. Use of AI-powered forecasting models to make predictions about future supply and demand based on the collected data.
4. Use of Simulation techniques to model the behaviour of the supply chain under different scenarios (i.e. changes in demand, supply disruptions, and blocked routes).
5. Use of optimisation algorithms to determine the optimal allocation of resources and to minimise costs in the supply chain.
6. Derive assumptions from the tool about the future state of the supply chain (i.e. demand trends, market structures, and the availability of suppliers).

Methodology Concept



Impact

MaaSiveTwin aims to make a substantial impact by addressing challenges in the CRM value chain, fostering a more efficient, resilient, and sustainable industrial ecosystem.

1. **Supply Chain Optimization**
 MaaSiveTwin optimizes the CRM value chain, facilitating informed decision-making through comprehensive data and analysis. This involves adjusting production schedules, mitigating supply chain disruptions, and identifying alternative supply sources, resulting in lower logistic costs, higher utilization rates, and reduced variable costs.
2. **Resilience and Sustainability**
 MaaSiveTwin improves industrial resilience and sustainability by offering insights into CRM demand, enabling swift adaptation to external conditions, facilitates the identification of mitigation strategies in case of threats, and promoting sustainable management practices and policy-making for environmental protection and sustainability.
3. **Global Leadership and Collaboration**
 MaaSiveTwin supports EU digitalization by promoting cross-cultural collaboration, enhancing IT skills, and raising awareness of sustainable practices. This collaboration fosters IT proficiency and critical thinking, positioning Europe as a leader in human-centric digital systems and infrastructure.



MaaSiveTwin

MANUFACTURING AS A SERVICE AND SUPPLY CHAIN PREDICTIVE TWIN FOR CRITICAL RAW MATERIALS

THANK YOU FOR YOUR ATTENTION!

Batronics
 FIRMA BATRONICS (Coordinator)
 TUHH
 TECHNISCHE UNIVERSITÄT HAMBURG
 RockTech
 Lithium
 ElevenEs
 RTO TALOS

Project Funded by
 European Union
 The work has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 101019719

Annexe 8 – Events, Journals, Magazines lists

Title	Category	Starting Date	Ending Date	Location	Interested Partner	Comments	Link (website)
The Battery Show Europe	Industrial event	3-Jun-25	5-Jun-25	Stuttgart, Germany		Access to the latest technologies from 1,100+ battery manufacturers across the industry supply chain. Explore the latest market innovations across Europe's largest battery technology expo.	https://www.thebatteryshow.eu/en/home.html
Battery Experts Forum	Industrial event	5-Nov-24	7-Nov-24	Darmstadt, Germany		The BATTERY EXPERTS FORUM is one of the largest international conferences and trade fairs for battery technology. Over 100 experts from leading battery manufacturers and users provide information on the latest trends, product developments and solutions.	https://www.battery-experts-forum.com/
Battery Innovation Days 2024	Scientific event	26-Nov-24	27-Nov-24	Barcelona, Spain or Online		Key European Research & Innovation Initiatives (Batteries Europe, Battery 2030+ and the Batteries European Partnership Association, in partnership with the Batteries 1st and 2nd IPCEIs are organising the BID in Barcelona and digitally on 26 – 27 November 2024. Battery Innovation Days are where leading policy makers, scientists and industry representatives discuss the future of battery R&I in Europe. BID aims to increase knowledge and encourage exchange around the deployment of cutting-edge technologies in battery materials, cell design, manufacturing and recycling.	https://battery2030.eu/news/happenings-events/battery-innovation-days-2024/
Battery Technologies Summit	Industrial event	25-Sep-24	26-Sep-24	Gebze, Turkey		This summit is bringing together leading names in the industry to share innovative ideas, present the latest technological developments and international perspectives. The summit will cover a wide range of topics covering every stage of the battery ecosystem, from raw material processes to cell production techniques, from extraordinary battery applications to sustainable waste strategies.	https://bepassociation.eu/battery-technologies-summit-2024/
Raw Materials Week 2024	Industrial event	9-Dec-24	13-Dec-24	Brussels, Belgium		The Raw Materials Week is an annual event organised by the European Commission. It gathers a wide range of stakeholders discussing policies and initiatives in the field of raw material, providing an overview of ongoing EU activities in this sector. It will focus on the first year of implementing the CRM Act, including the strategic projects and partnerships. The annual High-Level Conference on raw materials will take place on 11 December.	https://single-market-economy.ec.europa.eu/sectors/raw-materials/week_en
CRM for a Sustainable Future	Industrial event	7-Oct-24	9-Oct-24	Tallinn, Estonia		Geographically focusing on Northern Europe and the Baltic states, the conference will delve into various aspects of the critical raw material value chain. The theme encompasses geology, new exploration techniques, and environmental, social, and governance considerations on CRM.	https://eurogeosurveys.org/event/crm-for-a-sustainable-future/
IRTC25: From Raw Material Policies to Practice	Industrial event	19-Feb-25	21-Feb-25	Ljubljana, Slovenia		Explore how policies translate into real-world practices at IRTC25 in Ljubljana, where researchers, industry players, policymakers, and thought leaders convene to delve into the latest innovations and strategies shaping the future of critical raw materials. Over three dynamic days, this event will gather participants from around the globe to explore the most pressing issues and new advancements in critical raw materials. Abstract submission is open until October 13.	https://irtc.info/irtc-2025/
Raw Materials Summit	Industrial event	13-May-25	15-May-25	Brussels, Belgium		The Raw Materials Summit is a platform for industry leaders to connect, generate ideas and make change happen. Addressing the latest in technology, policy, sourcing, recycling, skills, circular economies and more, our speaker programme offers fresh perspectives on the industry - both upstream and downstream – past, present and future.	https://eltrmsummit.com/
Supply Chain Digitalization Conference & Exhibition 2025	Industrial event	1-May-25	2-May-25	Houston, TX, USA		The Supply Chain Digitalization Conference 2025 brings together supply chain and technology innovators to push forward a complete digital transformation within the supply chain. This platform will include the entire Supply Chain and Procurement, Transportation and Logistics industry together with Manufacturer, Retail, Consumer Goods, Chemical, Oil & Gas, Food & Beverage, Pharmaceuticals, Govt. Officials, Transportation & Logistics providers, and many more to discuss the latest developments and best practices in the industry.	https://supplychain-conference.com/
European Procurement & Supply Chain Excellence Summit 2025	Industrial event	31-Mar-25	1-Apr-25	Frankfurt, Germany		Explore strategies for driving purposeful, sustainable, and impactful change. Maximize your time and investment with discussions on: Macroeconomics & Geopolitics, Procurement Model of the Future, New Sources for Value Creation, Leadership, Talent & Culture, Sustainability as a Standard, Digitalization & AI in the Value Chain.	https://www.bme.de/veranstaltungen/grossveranstaltungen/european-procurement-supply-chain-excellence-summit-2025/
Supply Chain Visibility Conference & Exhibition 2025	Industrial event	6-May-25	7-May-25	Houston, TX, USA		The Supply Chain Visibility Conference 2025 will be the key to accelerate the strategic roadmap and hold a promising meeting place for the industry to develop new projects involving revolutionary technology using AI, IoT, Cloud computing and GPS enable real-time tracking and monitoring of goods throughout the supply chain, providing valuable data on location, condition, and performance.	https://visibility-conference.com/
Data Driven Supply Chain Conference 2025	Industrial event	30-Apr-25	30-Apr-25	Houston, TX, USA		An exclusive virtual event consisting of top-level content and thought leadership discussions exploring Supply Chain Visibility & Digital Revolution.	https://ptnevents.com/conferences/ddsc2025/
Sustainable Supply Chain Exhibition	Industrial event	25-Mar-25	26-Mar-25	Birmingham, UK		As environmental issues become more important to consumer and business brand owners across all sectors, there is growing demand for supply chains to adopt sustainable practices, become NetZero, and adopt effective ESG strategies. The Sustainable Supply Chain Exhibition will offer a focused environment for logistics, supply chain, and sustainability professionals to share ideas, offer solutions, and help each other on their sustainable journeys.	https://www.sustainableupplychainexhibition.co.uk/
Digital Twins Conferences 2024/2025/2026	Scientific event					Digital Twins Conferences from 2024-2026	https://conferenceindex.org/conferences/digital-twins

Figure 22: Comprehensive list of scientific or industrial events (conferences, exhibitions, fairs, etc.) of interest to MaaSiveTwin

Title	Category	Publisher	Comments	Interested Partners	ISSN or eISSN	Peer-reviewed	Open-access options	Link (website)
The Electrochemical Society	Scientific journal	Institute of Physics	The Electrochemical Society Interface is an authoritative yet accessible publication for those in the field of solid-state and electrochemical science and technology. Published quarterly, this full-color magazine contains technical articles about the latest developments in the field, and presents news and information about the Society.		1944-8783	TRUE	TRUE	https://iopscience.iop.org/journal/1944-8783
Electrochimica Acta	Scientific journal	Elsevier	The official journal of the International Society of Electrochemistry (ISE) Electrochimica Acta is an international journal publishing the highest quality original work and reviews in the field of electrochemistry and electrochemical engineering. This encompasses experimental, theoretical, computational and artificial intelligence or machine learning studies for which the primary focus is advancement of electrochemistry.		1873-3859	TRUE	TRUE	https://www.sciencedirect.com/journal/electrochimica-acta
CHEManager	Industrial magazine	Wiley	CHEManager International supplies managers and executives periodically with important information to assist them in making strategic business and investment decisions. Insightful interviews, substantiated market reports and technical articles provide 50,000 readers in Europe and overseas with a headstart on crucial business information and expert opinions 4 times per year.		0947-4188		TRUE	https://www.chemanager-online.com/en/chemanager-international
AutoMotorSport	Industrial magazine	Motor Presse Stuttgart	Auto Motor und Sport stands for automotive passion, spirit of discovery as well as independent and unique automotive journalism.		0005-0806		TRUE	https://www.auto-motor-und-sport.de/
Euractiv	Industrial magazine	Mediahuis	Euractiv is an independent pan-European media network specialised in EU affairs. Euractiv Media Network BV is owned by the European media publisher Mediahuis which is based in the Netherlands.					https://www.euractiv.com/
Journal of Applied Physics	Scientific journal	AIP Publishing	The Journal of Applied Physics is an influential international journal publishing full-length Articles, new or improved Methods, comprehensive Reviews, forward-looking Perspectives, and educational Tutorials. Topics covered in the journal are diverse and reflect the most current applied physics research, including: Magnetism, spintronics, and superconductivity		1089-7550	TRUE	TRUE	https://pubs.aip.org/aip/jap
IEEE Transactions on Industrial Electronics	Scientific journal	IEEE	Publishes papers with experimentally verified applications of electronics, controls, instrumentation, and computation for the enhancement of industrial systems and processes. Included are power electronics and electric motor drives, system control, signal processing, fault detection and diagnosis, power electronics in renewable energy and power systems, instrumentation, measurement and testing, modeling and simulation, motion control, applications in mechatronics and robotics, sensors and actuators, and applications of artificial intelligence in industrial electronic systems.		1557-9948	TRUE	TRUE	https://ieeexplore.ieee.org/xpl/RecentIssue.jspx?punumber=41
Journal of Supply Chain Management (JSCM)	Scientific journal	Wiley	Journal of Supply Chain Management (JSCM) is an international empirical journal known for its high-quality, high-impact research in the discipline of supply chain management. We welcome interdisciplinary research that employs qualitative or quantitative methods to develop, advance, or test theories, present novel interpretations, or challenge existing assumptions about SCM phenomena.		1745-493X	TRUE	TRUE	https://onlinelibrary.wiley.com/journal/1745493X
Supply Chain Management	Scientific journal	Emerald	The Journal of Supply Chain Management Science (JSCMS) is a peer-reviewed open-access journal (with no publishing fees) that aims to publish high-quality papers on quantitative approaches to supply chain management.		1359-8546	TRUE	TRUE	https://journals.open.tudelft.nl/jscms/index
Journal of Operations Management	Scientific journal	Wiley	The Journal of Operations Management (JOM) is one of the leading journals in the ISI Operations Research and Management Science category. JOM's mission is to publish original, empirical, operations and supply chain management research that demonstrates both academic and practical relevance.		1873-1317	TRUE	TRUE	https://onlinelibrary.wiley.com/journal/18731317
Journal of Purchasing and Supply Management	Scientific journal	Elsevier	The Journal covers the acquisition of external resources (materials, services, works, knowledge, etc.) by organizations in any sector (private, public, profit, non-profit), focusing in particular on strategy, design, management and evaluation of procurement and sourcing activities.		1478-4092	TRUE	TRUE	https://www.sciencedirect.com/journal/journal-of-purchasing-and-supply-management
International Journal of Physical Distribution & Logistics Management	Scientific journal	Emerald	International Journal of Physical Distribution & Logistics Management is a peer-reviewed academic journal which focuses on business logistics, supply chain management, physical distribution, and marketing channels.		1758-664X	TRUE	TRUE	https://www.emerald.com/insight/publication/issn/0960-0035
International Journal of Operations & Production Management	Scientific journal	Emerald	The International Journal of Operations and Production Management is a monthly peer-reviewed academic journal covering all aspects of supply chain management and operations management.		0144-3577	TRUE	TRUE	https://www.emerald.com/insight/publication/issn/0144-3577
Production and Operations Management	Scientific journal	Sage	The mission of Production and Operations Management is to serve as the flagship research journal in operations management in manufacturing and services. The journal publishes scientific research into the problems, interest, and concerns of managers who manage product and process design, operations, and supply chains.		1937-5956	TRUE	TRUE	https://journals.sagepub.com/home/paoa
Sustainable Materials and Technologies	Scientific journal	Elsevier	SM&T publishes original full-length research articles in applied or fundamental science of nano-, micro-, meso-, and macro-scale aspects of materials and technologies for sustainable development with special attention to contributions that reduce the knowledge gap between materials and system designs. Reviews that address significant challenges in the field and critical reviews which clearly address and limit the knowledge gap, provide a good guidance or real reference to the readers can be submitted.		2214-9937	TRUE	TRUE	https://www.sciencedirect.com/journal/sustainable-materials-and-technologies
Resources, Conservation and Recycling	Scientific journal	Elsevier	The journal Resources, Conservation & Recycling welcomes contributions from research, which consider sustainable management and conservation of resources. The journal emphasizes the transformation processes involved in a transition toward more sustainable production and consumption systems.		1879-0658	TRUE	TRUE	https://www.sciencedirect.com/journal/resources-conservation-and-recycling

Figure 23: Comprehensive list of Scientific Journals and Industrial Magazines of interest to MaaSiveTwin

Annexe 9 – Dissemination, Communication and Publications reporting

New item

Communication activity name
 Enter value here

Target audience *
 —

Who?

Communication channel *
 —

How?

Description *
 Enter value here

Describe the activity

Date *
 Enter a date

When did the activity took place?

Outcome *
 Enter value here

Status *
 Delivered

Partner *
 —

Select the partner(s) that participated in the dissemination/communication activity

Fullname *

Name the individuals that attended the event (Full name and Profession, for each name. One name per line) (e.g. John Smith, CEO Jane Doe, Assistant Professor etc)

Location *
 Enter value here

Reported on website/SoME
☐ Yes

Reported on SyGMA *
☐ Yes

Was this activity reported on the EC reporting platform?

Related URL
 Enter a URL

Alternative text (optional)

Provide a link for this activity (website news article, event agenda/details)

Image
 Add an image

Attachments
 Add attachments

Figure 24: Communication reporting via SharePoint - new activity form

New item

Dissemination activity name
Enter value here

Type of Activity *
—
What? Please select the category that best describes your activity type.

Target audience *
—
Who? Target audience Reached

Description *
Enter value here
Why? Description of the objective(s) with reference to a specific project output (max 200 characters)

Status *
Ongoing
Status of the dissemination activity

Partner *
—
Select the partner(s) that participated in the dissemination/communication activity

Date *
Enter a date
When did the activity took place?

Location *
Enter value here

Fullname *
Name the individuals that attended the event (Full name and Profession, for each name, One name per line) (e.g. John Smith, CEO Jane Doe, Assistant Professor etc.)

Related URL
Enter a URL
Alternative text (optional)
Link for event website/agenda etc.

Reported on website/SM
☐ Yes
Was this activity reported on the project's website and social media?

Reported on SyGMa
☐ Yes
Was this activity entered in the reporting platform of the EC

Image
Add an image

Image
Add an image

Attachments
Add attachments

Figure 25: Dissemination reporting via SharePoint - new activity form

New item

☐ Title of the scientific publication

Enter value here

☐ Authors

Enter value here

☐ Type of publication

—

☐ Title of the Journal or equivalent

Enter value here

☐ Month and Year of publication

Enter a date 

☐ Type of PID (repository)

—

☐ PID (publisher version of record)

Enter value here

☐ Link to publication

Enter a URL

Alternative text (optional)

☐ Publisher

Enter value here

☐ ISSN or eISSN

Enter value here

☐ Open access at the time of publication

☐ Yes

☐ Peer-reviewed

☐ Yes

☐ Charged OA publishing fees to the project

☐ Yes

☐ PID of deposited publication

Enter value here

☐ Number

Enter value here

☐ PID or Book

Enter value here

☐ Book title

Enter value here

☐ Processing costs to be charged on the project

Enter value here

☐ Partner

—

 Attachments

Add attachments

Figure 26: Publications reporting via SharePoint - new publication form